
An Ensemble Approach: Stacked Pre-Trained Deep Learning Bases and Logistic Regression Meta-Learner for Breast Cancer Histopathology Analysis.

**A thesis submitted in fulfilment of the requirements for the degree of
Bachelor of Science.**

By

RAKIBUR RAHMAN



**Department of Computer Science and Engineering
NATIONAL UNIVERSITY BANGLADESH 2022**

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Approval

This is the thesis titled “**An Ensemble Approach: Stacked Pre-Trained Deep Learning Bases and Logistic Regression Meta-Learner for Breast Cancer Histopathology Analysis**” submitted by **RAKIBUR RAHMAN** to the Department of Computer Science and Engineering, Institute of Science Trade and Technology, has been accepted as satisfactory for the fulfilment of the requirement for the degree of Bachelor of science in Computer Science and Engineering and approved as to its style and contents.

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DECLARATION

This is to certify that the thesis titled “**An Ensemble Approach: Stacked Pre-Trained Deep Learning Bases and Logistic Regression Meta-Learner for Breast Cancer Histopathology Analysis**” has been authored by me.

It presents the research conducted by me under the supervision of Shekh Tanjil Sharif. To the best of my knowledge, it is an original work, both in terms of research content and narrative, and has not been submitted elsewhere, in part or in full, for a degree.

Further, due credit has been attributed to the relevant state-of-the-art collaborations with appropriate citations and acknowledgements in line with established norms and practices.

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Bismillah al-Rahman al-Rahim

"In the name of Allah, the Most Gracious, the Most Merciful".

You have a plan; I have a plan. But there is also Allah's plan, which always overlaps our plan and proves that we are wrong, and Allah's plan is always better than ours.

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ABSTRACT

One of the most common and deadly types of cancer that affects women globally is breast cancer. Accurate diagnosis and early detection are essential for enhancing patient outcomes. Recent developments in deep learning and computer vision have demonstrated promising outcomes in the processing of medical images. To improve the precision of breast cancer detection, this thesis offers a study on the categorization of breast cancer images using Ensemble learning techniques with pre-trained deep learning model and traditional machine learning algorithm as meta model. In terms of classification accuracy and reliable feature extraction, this thesis showed Ensemble model to perform better than single deep learning model, highlighting its higher potential for precise breast cancer prediction from histography. The significance of Ensemble learning architectures like used propose model in enhancing medical picture classification tasks is highlighted by these findings.

Keyword: Deep Learning (DL), Ensemble Learning, Stacking, Ensemble, Pre-Trained Models, Transfer Learning, Logistic Regression Meta-Learner, Image Classification, Convolutional Neural Networks (CNN), Feature Extraction, Deep Learning in Healthcare.

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CHAPTER 01: Introduction

1.1 Background

By Global Cancer Observatory, Breast cancer is the *most prevalent* form of cancer, accounting for **2,261,419** (11.7%) of the **19,292,789** cases reported worldwide in 2020 [1].

Approximately 1 in 8 women in the United States (12.5%) will develop invasive breast cancer during their lifetime. In 2023, it's estimated that about 297,790 new cases of invasive breast cancer will be diagnosed in women in the US [2]. Kieffer et al. proposed a robust ensemble of deep learning models to address the variability of histopathology images. This ensemble learning approach performed ill across different test sets and magnification levels, demonstrating robustness in breast cancer classification.[3]

Zhu et al. used an ensemble of attention-guided CNN models to classify breast cancer histopathology images. By focusing on relevant regions within the tissue, the ensemble achieved higher sensitivity and specificity.[4]

Han et al. combined CNN features with SVM classifiers and further improved performance by employing ensemble learning techniques. Their method outperformed the traditional CNN-only approach, especially in differentiating between benign and malignant cases.[5]

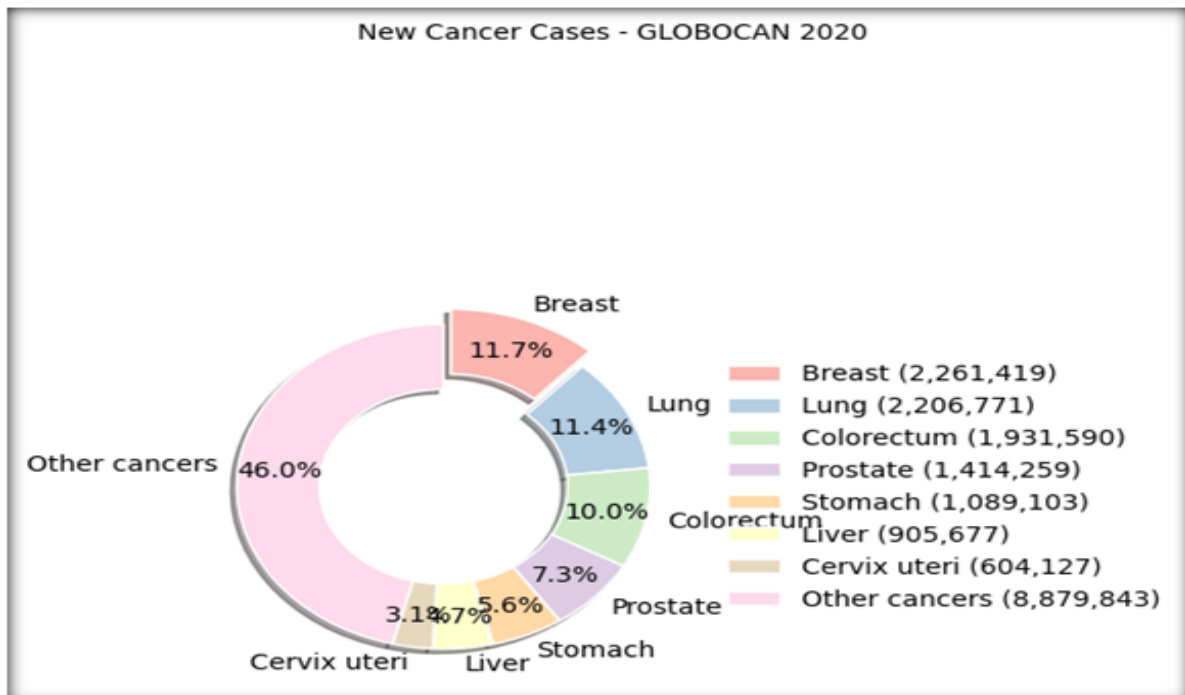


Figure 1.1 Cancer Statistics

Shallu et al. proposed a multi-scale ensemble approach for breast cancer classification by combining predictions from CNN models trained at different magnification levels (40X, 100X, 200X, 400X) on the BreakHis dataset. This approach yielded significant improvements in accuracy, particularly for high-magnification images.[6]

Ensemble models require large amounts of annotated data for training, often involving sensitive patient information. Privacy concerns around sharing histopathological data, especially across borders, pose a challenge to developing large, diverse datasets that can improve the generalization of ensemble models.

Pre-trained model needs large amount of data for training. If using small amount of data set for train this kind of model then these will be over fitting. So, I need to augmentation dataset.

Ensemble models, particularly those that involve deep learning, are computationally expensive due to the need to train and run multiple models simultaneously. This increases inference time, making them less feasible for real-time clinical applications where quick diagnoses are critical. Estevan et al. mentions the trade-off between accuracy and efficiency in deploying ensemble models in clinical settings. They stress the need for lighter and faster models without compromising diagnostic performance.[7]

Breast cancer histopathology datasets often exhibit class imbalance, where certain types of cancer or benign samples are overrepresented. This can bias the model towards the majority class, reducing its effectiveness in detecting rarer, but critical, cases. Existing ensemble methods have not sufficiently addressed this issue, and it remains a challenge. Sun et al. discussed the problem of class imbalance in medical image datasets and argue that ensemble models, though more robust than individual models, can still struggle with minority class detection without proper data augmentation or resampling techniques.[8]

1.2 Motivation

Clinicians can use the richness of information in histopathology images to help them make precise diagnostic and prognostic judgments. However, analyzing these images requires a pathologist's knowledge and can be difficult and time-consuming. The goal of our research is to use machine learning to create reliable and accurate categorization models that can assist pathologists in their evaluations. I am attempting to classify images of breast cancer histopathology with the goal of providing clinicians with cutting-edge tools that not only speed up analysis but also improve accuracy.

Such technology has a wide range of possible advantages, from assisting in early detection to forecasting disease aggression, which could ultimately result in more specialized and efficient treatment approaches. Our drive is also fueled by the chance to close the gap between medical knowledge and technological advancement. Data scientists and medical practitioners working together can create ground-breaking solutions that directly improve patient care. This research aims to create a synergy between human knowledge and artificial intelligence as trained models to recognize patterns and subtleties in histopathological images.

1.3 Problem Statement

Breast cancer is one of the most prevalent forms of cancer worldwide, and early and accurate diagnosis is crucial for effective treatment and patient survival. Histopathological analysis of tissue samples is a standard method for diagnosing breast cancer, where pathologists examine tissue under a microscope to detect the presence of malignancy. However, manual interpretation of histopathological images is time-consuming, prone to human error, and highly dependent on the experience of the pathologist.

Advances in deep learning, particularly Convolutional Neural Networks (CNNs), have shown significant potential in automating the analysis of medical images, including histopathological slides. However, despite these advances, individual deep learning models often suffer from limitations such as overfitting, sensitivity to noise, and difficulty generalizing across different datasets. These shortcomings are particularly critical in medical image analysis, where accuracy and reliability are paramount.

Ensemble learning, which combines the predictions of multiple models, has been shown to improve accuracy and robustness in machine learning tasks. Yet, there is limited research on applying ensemble techniques, such as stacking, to deep learning models in the context of breast cancer histopathology. Furthermore, the use of logistic regression as a meta-learner to combine the outputs of pre-trained CNN models has not been fully explored.

Therefore, the problem this thesis addresses is the development of an ensemble framework that combines multiple pre-trained deep learning models through stacking, using logistic regression as a meta-learner, to enhance the accuracy, reliability, and generalizability of breast cancer histopathology analysis. The challenge lies in determining whether this ensemble approach can outperform individual models and existing state-of-the-art methods in this domain, as well as its applicability to real-world clinical settings.

1.4 Research Objective

- Comparative analysis of using proposed method for breast cancer histopathology image classification.
- Comparison Between Single pre-trained model with Staked architecture.
- To SLN assessment's challenges are Illustrated this demonstrates the need for advancements in the detection and treatment of cancer.
- To deep learning algorithms for Proposing a Solution using the Stacked architecture. The specific objectives of this study are as follows.
- To data preparation and augmentation with the Stacked architecture and enhance the model's ability to generalize.

- To implement An Ensemble Approach to accommodate the specific classification task and the number of classes involved.
- To transfer learning and pretraining base MobileNet and EfficientNetB0 on the breast cancer histopathology images dataset.
- Hyperparameter Tuning and Optimization for mitigate overfitting and ensure the model's generalization capability.
- Model Evaluation and Performance Metrics.
- Training Meta Learner Logistic Regression model and make final Classification.
- Interpretability and Visualization.

1.5 Structure of the Thesis

In Chapter 1: Briefly introduced about breast cancer, scope and limitation, Research Objective.

In Chapter 2: Remaining sections are organized as follows: Related study, Digital Image and Medical images, Artificial Intelligence, Deep Learning, Pre-trained model, Transfer Learning, Ensemble Approaches.

In Chapter 3: The approach is described in full, including information on data collection, preprocessing, and the fine-tuning base models MobileNet, EfficientNetB0 model and proposed method design.

In Chapter 4: Explain Result and Analysis, Model evaluation metrics, Performance and Comparative Analysis.

In Chapter 5: A thorough study of the outcomes and Future study and Recommendations also limitation

Chapter 2: Literature Review

2.1 Related Studies on Breast Cancer Classification

Certainly, here are a few studies that focus on breast cancer classification using various techniques, including deep learning, machine learning, and other image analysis methods:

This review paper provides an extensive overview of the recent advances in applying deep learning techniques to breast cancer classification using histopathological images. It covers various deep learning architectures, data augmentation strategies, and challenges in the field.[9]

This study investigates the use of convolutional neural networks (CNNs) for classifying breast cancer histology images. It explores different CNN architectures and evaluates their performance on a publicly available dataset.[10]

This review paper provides an in-depth analysis of various image analysis techniques applied to breast cancer classification using histopathological images. It covers traditional image analysis methods as well as machine learning and deep learning approaches.[11]

Similar to the previous study, this paper provides a comprehensive review of the methodologies employed in breast cancer histopathological image classification. It covers feature extraction techniques, machine learning algorithms, and deep learning models.[12]

This study focuses on using deep neural networks for breast cancer histopathology image classification. It compares different neural network architectures and discusses the challenges and opportunities in the field.[13]

This study explores the use of thermal images and machine learning techniques for breast cancer classification. It demonstrates the potential of non-invasive methods for early cancer detection.[14]

This study performs a comparative analysis of different machine learning algorithms for breast cancer histopathology image classification. It evaluates the performance of traditional and deep learning models on the dataset.[15]

This research explores the use of ensemble techniques to combine multiple deep learning models for improved breast cancer histology image classification accuracy. These studies represent a mix of approaches, including deep learning, traditional machine learning, and a variety of imaging modalities. They provide insights into the state-of-the-art Methods and advancements in breast cancer classification using medical images.[16]

Overview of existing approaches for breast cancer prediction:

S.No. Author(s)	Approach	Dataset	Performance Measures
1. Jabbar et al. (2021) [17]	Bayesian network, Radial Basis Function.	Wisconsin breast Cancer Dataset (WBCD)	Accuracy-97%
2. Anastraj et al. (2019) [18]	Back propagation Network, ANN, CNN, SVM	Wisconsin breast Cancer (original) dataset	Accuracy-94%
3. Kasaudhan et al. (2015) [19]	SVM	Digital Database For Screening Mammography (DDSM)	Accuracy-91.5%
4. Mejia et al. (2015) [20]	KNN	Federal Fluminense University Hospital	Accuracy-94.44%
5. Jiang, and XU (2017) [21]	RF-Recursive Feature Elimination method	Zhejiang Cancer Hospital	Accuracy-77.05%
6. Bevilacqua et al. (2019) [22]	Optimised ANN	Radiologists of the University of Bari Aldo Moro	Accuracy-89.77%
7. Nanglia et al. (2022) [23]	Heterogeneous ensemble model	Anthropometric data	Accuracy-78%
8. Lamba et al. (2023) [24]	Consistency-first Search-Best First search (CFS-BFS) and Consistency-Best First Search (Consistency-BFS)	CFS-BFS and Consistency-BFS	Accuracy—66% Logistic Regression
9. Li et al. (2023) [25]	An ensemble classifier based on multilayer perceptron.	Wisconsin Database	Accuracy—91%
10. Al lmanet al. (2023) [26]	Classification performance was compared between Gail and BRISK.	Nested case-control dataset	(NRI = 0.287; SE = 0.035).

Overview of papers dealing with various methods and findings based on BreakHis Dataset:

Authors	Dataset	Methods	Findings	Accuracy
Myung [27]	BreakHis	Transfer Learning, Ddata-Im balance-under sampling, applied on VGG16 and InceptionV3	Binary Classification	98%
Dalal BarDau [28]	BreakHis	Extraction of a set of hand-crafted features was encoded by using two models, Bag-of words and SVM	Binary Classification	96.15% - 98.33%
Neslihan Bayromogu [29]	BreakHis	Designed Own CNN architecture for classification.	Magnification Benign and Malignant	77.3± 5.91% - 83±8.54% 82.1±4.% - 83.1±3.5%
F.A. Spanhol [30]	BreakHis	Used AlexNet for Classification	Accuracy	90.00%± 6.7
Majid Nawaz [31]	BreakHis	Compared the results with LeNet, AlexNet and DenseNet.	Accuracy	95.4%
Zhongyi Han [32]	BreakHis	Compared the LeNet, AlexNet and CSDCNN with Raw Data and Augmented Data.	Accuracy	93.2%
Abdullah-Al Nahid [33]	BreakHis	Compared CNN, LSTM and CNN-LSTM Architecture using both softmax layer and SVM Classifier	Accuracy	91.00%
M. Jannesari [34]	TMA Database and BreakHis	Examined Different neural Networks such as Inception (V1, V2, V3 and V4) and ResNet (V1 50, V1 101 and v1 152)	Accuracy	98.4%

2.2 Research Gap

Identifying research gaps is crucial for advancing the field of breast cancer classification using Convolutional Neural Networks (CNNs). Despite significant progress, several areas require further investigation to enhance the accuracy, reliability, and clinical applicability of CNN-based models. Below is a comprehensive overview of the current research gaps in this domain.

2.2.1 Data Related Gaps

Limited and Imbalanced Datasets:

- **Scarcity of Large-Scale Datasets:** High-quality, annotated datasets for breast cancer imaging (e.g., histopathology, mammography) are limited. Access to large and diverse datasets is essential for training robust CNN models.
- **Class Imbalance:** Many datasets suffer from imbalanced classes, where malignant cases are fewer than benign ones, leading to biased models that may underperform on minority classes.

Data Quality and Variability:

- **Image Quality and Standardization:** Variations in imaging protocols, resolution, and quality across different sources can affect model performance. Standardization is often lacking, making it difficult for models to generalize.
- **Labeling Consistency:** Inconsistent labeling and annotation practices across datasets can introduce noise, reducing the effectiveness of supervised learning approaches.

Lack of Multi-Model Data Integration:

- **Limited Use of Multi-Modal Data:** Most studies focus on a single imaging modality (e.g., mammograms) rather than integrating multiple sources of data (e.g., MRI, ultrasound, genetic information), which could provide a more comprehensive understanding of breast cancer.

2.2.2 Model Related Gaps

Interpretability and Explain ability:

- **Black-Box Nature of CNNs:** CNN models often lack transparency, making it difficult for clinicians to understand the decision-making process, which is critical for clinical trust and adoption.
- **Limited Explainable AI (XAI) Techniques:** There is a need for more advanced XAI methods tailored to breast cancer classification to provide meaningful insights into model predictions.

Overfitting and Generalization:

- **Overfitting to Training Data:** High-capacity CNNs can overfit to limited datasets, reducing their ability to generalize to unseen data.
- **Domain Adaptation:** Models trained on specific datasets may not perform well across different populations or imaging conditions, indicating a need for better domain adaptation techniques.

Transfer Learning Limitations:

- **Pretrained Models on Non-Medical Data:** Many CNN models leverage transfer learning from datasets like ImageNet, which may not capture the nuances of medical imaging, limiting their effectiveness.
- **Fine-Tuning Challenges:** Effective strategies for fine-tuning pretrained models on breast cancer data are still underexplored.

Computational Constraints:

- **Resource-Intensive Training:** Training deep CNNs requires significant computational resources, which can be a barrier for researchers with limited access to high-performance computing.
- **Inference Efficiency:** Deploying CNN models in clinical settings demands efficient inference to provide timely results, necessitating optimization for speed and resource usage.

2.2.3 Clinical Applicability Gaps

Integration with Clinical Workflows:

- **Lack of Seamless Integration:** There is limited research on integrating CNN-based classification tools into existing clinical workflows, which is essential for practical adoption.
- **User Interface and Experience:** Designing user-friendly interfaces that present model predictions and explanations effectively to clinicians remains a challenge.

Validation on Diverse Populations:

- **Limited Demographic Representation:** Models often lack validation across diverse populations, including different ethnicities, ages, and genetic backgrounds, which can affect their generalizability and equity.
- **Real-World Clinical Data:** Most studies use retrospective data; prospective validation using real-world clinical data is necessary to assess model performance in practical settings.

2.2.4 Novel Approaches and Unexplored Areas

Combining CNNs with Other Models:

- **Ensemble Methods:** Developing ensemble approaches that combine multiple models may improve classification accuracy and robustness.
- **Hybrid Models:** Integrating CNNs with other machine learning techniques could enhance performance and capture more complex patterns.

Incorporating Domain Knowledge:

- **Feature Engineering with Medical Insights:** Leveraging domain-specific knowledge to guide feature extraction and model architecture can enhance model interpretability and performance.
- **Pathology-Informed Models:** Incorporating insights from pathology, such as cellular structures and tissue patterns, can lead to more accurate and meaningful classifications.

Real-Time Analysis and Deployment:

- **On-Device Processing:** Research into deploying CNN models on edge devices for real-time analysis during clinical procedures is limited.
- **Scalability and Deployment Strategies:** Effective strategies for scaling and deploying models in diverse clinical environments need further exploration.

While CNNs have shown promise in improving breast cancer classification, several research gaps hinder their full potential in clinical applications. Addressing data limitations, enhancing model interpretability, ensuring generalization, integrating with clinical workflows, and establishing standardized evaluation protocols are critical steps for future research. By focusing on these areas, the field can move towards more accurate, reliable, and clinically applicable CNN-based breast cancer classification systems.

2.3 Image

An image is a visual representation of an object, scene, or concept, typically created through the capture or synthesis of light or other types of electromagnetic waves. It can be a two-dimensional (2D) or three-dimensional (3D) depiction that conveys information through variations in color, brightness, texture, or form. Images can be generated through a variety of methods, including photography, drawing, painting, or computational rendering.

Types of Images:

- **Physical Image:** A tangible representation, such as a printed photograph or painting.
- **Digital Image:** A numerical representation of an image, typically in the form of pixels, stored and displayed on computers or electronic devices.
- **Mental Image:** A conceptual image formed in the mind without the need for physical visual input.

2.3.1 Image and Digital Image Processing

Image processing is a field of computer science and engineering that involves the manipulation and analysis of digital images to enhance them, extract information, or transform them into a format suitable for further analysis. It encompasses a wide range of techniques and methods, often relying on algorithms and mathematical models to perform various operations on images.

Digital image processing is the use of computer algorithms to perform image manipulation, enhancement, and analysis on digital images. It involves transforming images to improve their quality, extract useful information, and facilitate tasks such as object recognition, feature extraction, and classification. Unlike analog image processing (which deals with visual or optical systems), digital image processing focuses on handling discrete numerical data representing the pixels of an image.

2.3.2 Fundamental steps of digital image processing

Image Acquisition: Obtaining or capturing an image via sensors (e.g., cameras, scanners).

Image Restoration: Recovering or correcting images that have been degraded (blur, noise, etc.)

Image Enhancement: Improving the visual appearance of the image by adjusting contrast, brightness, or reducing noise.

Image Restoration: Recovering or correcting images that have been degraded (blur, noise, etc.).

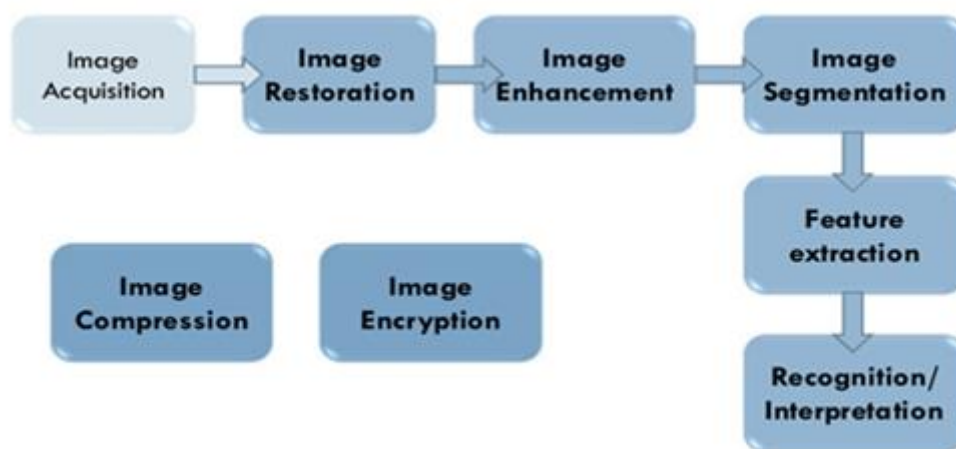


Figure 2.3.2: Steps of digital image processing.

Image Segmentation: Dividing an image into meaningful parts or regions for analysis.

Feature Extraction: Identifying and isolating important features such as edges, textures, or shapes.

Image Compression: Reducing the size of an image file without significant loss of quality.

Image Recognition: Identifying and classifying objects or patterns within the image using algorithms or AI.

Application of Digital Image Processing:

- **Medical Imaging:** MRI, CT scans, X-ray analysis.
- **Remote Sensing:** Satellite imagery for environmental monitoring or urban planning.
- **Machine Vision:** Autonomous vehicle navigation, industrial quality control.
- **Photography:** Noise reduction, color correction, and image enhancement.
- **Security:** Face recognition, biometric analysis, surveillance systems.

2.4 Artificial Intelligence

Artificial Intelligence (AI) is a branch of computer science focused on creating machines and software that can perform tasks that typically require human intelligence. These tasks include problem-solving, learning, reasoning, perception, and natural language understanding.

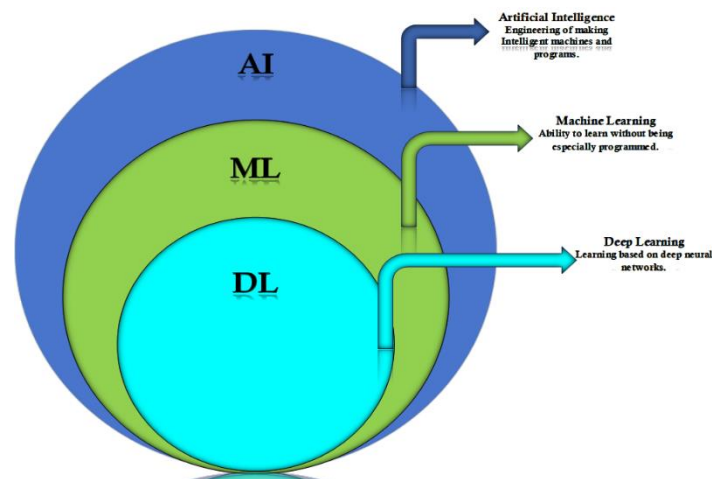


Figure 2.4: Relation Between AI, ML, DL

Key Concepts in AI:

- **Machine Learning (ML):** A subset of AI that enables systems to learn from data and improve over time without being explicitly programmed.
- **Neural Networks:** Algorithms inspired by the human brain that are used in deep learning, a powerful form of machine learning.
- **Natural Language Processing (NLP):** AI's ability to understand and generate human language.
- **Computer Vision:** Enabling machines to interpret and understand visual information from the world, like images or videos.
- **Robotics:** Using AI to control robots that can perform tasks autonomously or semi-autonomously.

2.4.1 Machine Learning

Machine Learning (ML) is a subset of artificial intelligence (AI) that enables systems to learn from data, identify patterns, and make decisions or predictions without explicit programming for each task. Instead of following predefined rules, machine learning algorithms adapt and improve over time as they are exposed to more data.

Types of Machine Learning:

- **Supervised Learning:** The model is trained on labeled data, where the input-output pairs are known. It learns to predict the output for new, unseen inputs.
Example: Spam detection in emails, where emails are labeled as spam or not spam.
- **Unsupervised Learning:** The model works with unlabeled data, discovering patterns, groupings, or structures in the data.
Example: Customer segmentation based on purchasing behavior, where no predefined labels exist.
- **Reinforcement Learning:** The model learns by interacting with an environment, receiving feedback in the form of rewards or penalties, and adjusting its actions accordingly.
Example: Training an AI to play a game, where it learns to win by trial and error.

Key Components of Machine Learning:

- **Data:** Raw information used to train the machine learning model. This data can be structured (like spreadsheets) or unstructured (like images or text).
- **Algorithms:** Set of rules or mathematical models used to learn from data and make predictions. Popular algorithms include decision trees, neural networks, and support vector machines.
- **Model:** The resulting system after an algorithm has learned from the data. It can make predictions or decisions based on new input data.
- **Training:** The process of feeding data to the algorithm so it can learn the patterns or features within the data.
- **Testing:** Evaluating the model on unseen data to measure its performance and accuracy.

2.4.2 Traditional Machine Learning Algorithm

Traditional machine learning algorithms are foundational methods used to build models that can learn from data and make predictions or decisions. These algorithms are typically categorized into supervised, unsupervised, and semi-supervised learning methods. Below are some of the most commonly used traditional machine learning algorithms:

- Linear Regression
- Logistic Regression
- Decision Trees
- Random Forest
- Support Vector Machines (SVM)

2.5 Deep Learning

Deep Learning is a subset of machine learning, which itself is a branch of artificial intelligence (AI). It involves the use of artificial neural networks with multiple layers (hence "deep") to model and understand complex patterns and representations in data. Here's a more detailed breakdown:

2.5.1 Key Concepts of Deep Learning

Artificial Neural Networks (ANNs):

- **Neurons:** The basic units inspired by biological neurons in the human brain. Each neuron receives input, processes it, and passes the output to the next layer.
- **Layers:**
 - Input Layer: Receives the raw data.
 - Hidden Layers: Intermediate layers where computations are performed. Deep learning models typically have multiple hidden layers.
 - Output Layer: Produces the final prediction or classification.

Deep Neural Networks (DNNs):

- **Depth:** Refers to the number of hidden layers in the network. More layers allow the model to learn more abstract and complex features.
- **Types of DNNs:**
 - Convolutional Neural Networks (CNNs): Primarily used for image and video recognition tasks.
 - Recurrent Neural Networks (RNNs): Suited for sequential data like time series or natural language.
 - Generative Adversarial Networks (GANs): Used for generating new data samples that resemble the training data.

Training Deep Learning Models:

- **Data:** Requires large amounts of labeled data to train effectively.
- **Backpropagation:** The algorithm used to calculate the gradient of the loss function and update the network's weights.
- **Optimization Algorithms:** Such as Stochastic Gradient Descent (SGD), Adam, and RMSprop, which adjust the weights to minimize the loss function.
- **Activation Functions:** Non-linear functions like ReLU, sigmoid, and tanh that introduce non-linearity into the model, enabling it to learn complex patterns.

Applications of Deep Learning:

- Computer Vision: Image classification, object detection, facial recognition, and image generation.
- Natural Language Processing (NLP): Language translation, sentiment analysis, chatbots, and text generation.
- Speech Recognition: Converting spoken language into text, virtual assistants.
- Healthcare: Medical image analysis, drug discovery, personalized medicine.
- Autonomous Vehicles: Perception, decision-making, and control systems in self-driving cars.
- Finance: Fraud detection, algorithmic trading, and risk management.

2.5.2 Convolutional Neural Network

Convolutional Neural Networks (CNNs) are a specialized type of artificial neural network designed to process structured grid data, such as images. They have revolutionized the field of computer vision and have applications in image and video recognition, recommendation systems, natural language processing, and more. Here's an in-depth look at CNNs:

A Convolutional Neural Network (CNN) is a deep learning model that automatically and adaptively learns spatial hierarchies of features from input data through backpropagation by using multiple building blocks, such as convolution layers, pooling layers, and fully connected layers.

Key Characteristics:

- Local Connectivity: Neurons in a layer are only connected to a small region of the input, preserving spatial relationships.
- weight Sharing: The same set of weights (filters) is used across different regions of the input, reducing the number of parameters and capturing translational invariance.
- Hierarchical Feature Learning: CNNs learn hierarchical patterns, starting from low-level features like edges and textures to high-level features like objects and faces.

A typical CNN architecture consists of several types of layers arranged in a specific order:

Input Layer:

- Accepts the raw input data, such as an image with dimensions (height, width, channels).
Example: A color image might have dimensions 224x224x3 (height x width x RGB channels).

Convolutional Layers:

- Convolution Operation: Applies multiple filters (kernels) to the input to extract features.
- Filters/Kernels: Small matrices (e.g., 3x3, 5x5) that slide over the input to detect specific features like edges, textures, or patterns.
- Stride: Determines the step size with which the filter moves across the input.

- **Padding:** Adds borders to the input to control the spatial size of the output.
- **Activation Function:** Typically uses non-linear functions like ReLU to introduce non-linearity.

Architecture of CNN:

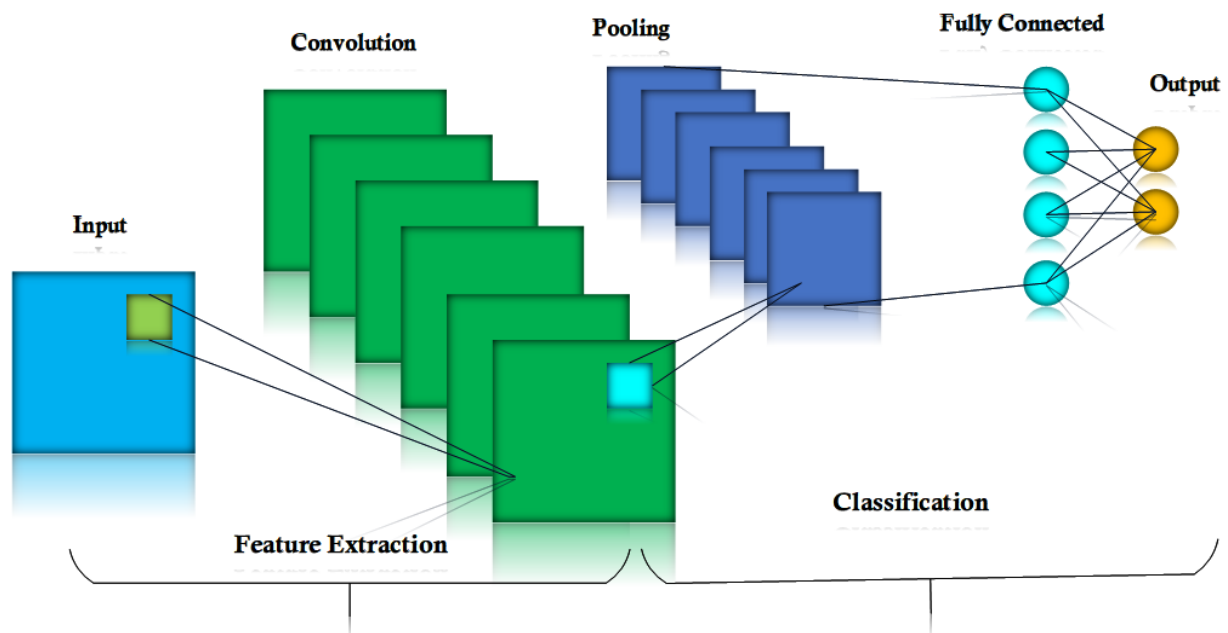


Figure 2.5.2: Basic Architecture of CNN

Formula:

$$\text{Output}(i,j) = \text{Activation}\left(\sum_m \sum_n \text{Input}(i+m, j+n) \times \text{Filter}(m, n) + \text{Bias}\right)$$

$$\text{Output}(i, j) = \text{Activation} \left(\sum_m \sum_n \text{Input}(i + m, j + n) \times \text{Filter}(m, n) + \text{Bias} \right)$$

Pooling Layers:

- **Purpose:** Reduce the spatial dimensions (height and width) of the feature maps, decreasing computational load and controlling overfitting.
- **Types:**
 - **Max Pooling:** Takes the maximum value from each window (e.g., 2x2).
 - **Average Pooling:** Computes the average value within each window.
 - **Stride and Window Size:** Determines how the pooling window moves across the input.

Fully Connected (Dense) Layers:

- Connect every neuron from the previous layer to every neuron in the current layer.
- Serve as a classifier based on the features extracted by convolutional and pooling layers.
- Typically found towards the end of the network.

Output Layer:

- Produces the final prediction, such as class probabilities in classification tasks.
- Uses activation functions like SoftMax for multi-class classification or Sigmoid for binary classification.

Working procedure of CNN

Feature Extraction:

- Convolutional Layers: Detect local patterns using filters.
- Activation Functions: Introduce non-linearity, allowing the network to learn complex patterns.
- Pooling Layers: Down sample the feature maps, making the representations smaller and more manageable.

Hierarchical Learning:

- Early layers capture simple features (e.g., edges).
- Intermediate layers capture more complex features (e.g., textures, shapes).
- Deeper layers capture high-level features (e.g., objects, faces).
- Classification:
- Fully Connected Layers: Integrate the learned features to perform the final classification or regression task.

Training:

- Forward Propagation: Input data passes through the network to generate predictions.
- Loss Calculation: Measures the difference between predictions and actual labels.
- Backpropagation: Computes gradients of the loss with respect to each weight.
- weight Updates: Adjusts the weights using optimization algorithms (e.g., SGD, Adam) to minimize the loss.

Key Components and Concepts

Filters/Kernels:

- Small-sized matrices that convolve with the input to produce feature maps.
- Each filter learns to detect a specific feature.

Stride:

- Controls how the filter moves across the input.
- Larger strides result in smaller output dimensions.

Padding:

- Valid Padding: No padding; output size decreases.
- Same Padding: Adds zeros to maintain the input size after convolution.

Activation Functions:

- ReLU (Rectified Linear Unit): Sets negative values to zero, introducing non-linearity.
- Sigmoid/Tanh: Used in certain layers for specific purposes.

Normalization:

- Batch Normalization: Stabilizes and accelerates training by normalizing layer inputs.

Dropout:

- A regularization technique that randomly deactivates neurons during training to prevent overfitting.

Popular CNN Architecture: Several groundbreaking CNN architectures have been developed, each introducing novel techniques to improve performance:

MobileNet:

- Designed for mobile and edge devices, MobileNet uses depth wise separable convolutions to reduce the number of parameters while maintaining performance.
- It's light weight and efficient for real-time applications.

EfficientNetB0:

- An extension of the Inception architecture, EfficientNetB0 uses depth wise separable convolutions, improving efficiency and performance on various tasks.

LeNet-5:

- Developed by Yann LeCun in the 1990s, it was one of the first CNN architectures and is primarily used for handwritten digit recognition (MNIST).
- Features include convolutional layers, subsampling (pooling) layers, and fully connected layers.

AlexNet:

- Introduced in 2012 by Alex Krizhevsky, it won the ImageNet Large Scale Visual Recognition Challenge (ILSVRC) that year.
- It has five convolutional layers, max pooling, and three fully connected layers, utilizing ReLU activations and dropout for regularization.

VGGNet:

- Developed by the Visual Geometry Group at Oxford, VGGNet (especially VGG16 and VGG19) is known for its simplicity and depth, using 3x3 convolutional filters.

- It features a very deep architecture (up to 19 layers) and employs max pooling after convolutional layers.

ResNet (Residual Networks):

- Developed by Microsoft Research, ResNet introduced skip connections to help with the vanishing gradient problem, allowing for very deep networks (up to 152 layers).
- It uses residual blocks, which add the input of a layer to the output of a subsequent layer.

DenseNet (Densely Connected Convolutional Networks):

- Similar to ResNet, but instead of adding inputs, it concatenates the outputs of all preceding layers to each subsequent layer, enhancing feature reuse.
- DenseNet architectures can also be very deep, improving gradient flow.

2.6 Breast Cancer and Imaging Modalities

Breast cancer is a type of cancer that originates in the cells of the breast. It is one of the most common cancers diagnosed in women, but it can also occur in men. Early detection and accurate diagnosis are crucial for effective treatment and improved outcomes. Imaging modalities play a significant role in both the detection and diagnosis of breast cancer. Breast cancer can be grouped in several ways, including by its histologic type, clinical characteristics, and expression of tumor markers. Ductal and lobular carcinomas account for roughly 75 and 15% of all invasive breast cancer cases in the US, respectively, and are the two most prevalent histologic categories.[12] Here are some commonly used imaging modalities for breast cancer:

Histopathology: Histopathology is the study of diseased tissues under a microscope, with the aim of understanding the manifestations and causes of diseases at the cellular level. It plays a crucial role in medical diagnosis, particularly in identifying cancer and other pathological conditions.

Mammography: Mammography is the most common and widely used imaging modality for breast cancer screening. It involves taking X-ray images of the breast tissue. Mammograms can detect abnormalities such as calcifications or masses in the breast tissue, often before they can be felt during a physical examination. Digital mammography has become the standard, allowing for better image manipulation and storage.

Ultrasound (Sonography): Breast ultrasound uses high-frequency sound waves to create images of the breast tissue. It is often used to further evaluate abnormalities found on -

mammograms or physical examinations. Ultrasound can differentiate between solid masses and fluid filled cysts, aiding in the diagnosis and characterization of breast lesions.

Magnetic Resonance Imaging (MRI): Breast MRI is a powerful imaging technique that uses a strong magnetic field and radio waves to create detailed images of the breast tissue. It is

particularly useful in certain situations, such as assessing the extent of disease in high-risk individuals and characterizing suspicious lesions found on mammograms or ultrasound.

Computed Tomography (CT) Scan: While not commonly used for routine breast imaging, CT scans can be used to assess the chest and surrounding areas for signs of metastasis (spread of cancer to other parts of the body). CT scans provide detailed cross-sectional images of the body.

Positron Emission Tomography (PET) Scan: PET scans can be used to detect areas of increased metabolic activity in the body, which can indicate the presence of cancer cells. In breast cancer, PET scans are often used to evaluate the extent of disease and identify potential metastases.

Molecular Breast Imaging (MBI): MBI is a specialized imaging technique that uses a small amount of radioactive material to create images of breast tissue. It is particularly useful in dense breast tissue, where mammography may be less effective at detecting abnormalities. Each imaging modality has its strengths and limitations, and the choice of modality depends on factors such as the patient's risk profile, the type of lesion being evaluated, and the purpose of the imaging (screening, diagnosis, staging, etc.). Often, a combination of different imaging techniques may be used to get a comprehensive view of the breast tissue and any potential cancerous lesions.

2.7 Deep Learning in Medical Image Analysis

Deep learning has achieved remarkable success in medical image analysis and has demonstrated its potential in various applications. Deep learning has the ability to organize and extract discriminative information from data without the need for a domain expert to construct feature extractors [8]. Its success is evident in the following area.

Disease Detection and Diagnosis: Disease detection and diagnosis Deep learning models have demonstrated great accuracy in identifying a variety of diseases from medical photos. They have been employed, for example, to detect malignancies (breast, lung, and skin), neurological problems (Alzheimer's, Parkinson's), cardiovascular conditions (heart disease, aneurysms), and more.

Localization and Segmentation: Deep learning has made it possible to precisely segment and locate structures of interest inside medical images. Planning treatments, operations, and interventions depends on this. Segmenting organs, tumors, blood vessels, and lesions are some examples.

Image classification: Deep learning algorithms can accurately classify medical photos into several categories. They can distinguish between various skin lesions, classify retinal images to identify diabetic retinopathy, and identify lung nodules in CT scans, among other things.

Radiomics and Predictive Modeling: Complex features from medical images are extracted using deep learning to enable the development of predictive models. Based on the study of

imaging data, these models can forecast disease development, treatment response, and patient outcomes.

Screening and early detection: Deep learning is useful for early disease detection and screening programs because of its capacity to recognize subtle abnormalities. Deep learning algorithms, for instance, assist mammography in identifying breast cancer in its early stages.

Image Reconstruction and Enhancement: Medical picture quality has been improved using deep learning approaches, which also reduce noise and artifacts. Additionally, they help in picture reconstruction from sparse or noisy data, increasing the diagnostic utility of subpar scans.

Personalization and customization: Deep learning models can be trained to identify traits unique to a patient and adjust treatment strategies accordingly. This method is very useful in precision medicine, where therapies are tailored to each patient's distinct profile.

Resource Optimization: By automating operations like image preparation, quality evaluation, and anomaly identification, deep learning can enhance the effectiveness medical imaging workflows while saving time and resources for medical personnel. Even though deep learning has had great success, there are still issues to be resolved. Large and diverse datasets are required; there are issues with model interpretation and generalization to new data; there may be biases in model predictions; there are legal issues to think about; and AI systems are now being integrated into clinical practice.

2.8 Pre-trained Model

A pretrained model is a machine learning model that has been previously trained on a large dataset and is made available for reuse in different tasks or applications. Here are some key characteristics and advantages of using pretrained models:

Key Characteristics

Training on Large Datasets:

- Pretrained models are typically trained on large and comprehensive datasets, such as ImageNet for image classification tasks. This extensive training allows them to learn rich feature representations.

Transfer Learning:

- Pretrained models can be fine-tuned for specific tasks. This process, known as transfer learning, involves adapting a model trained on one task to work effectively on a different but related task.

Layer Freezing:

- In fine-tuning, some layers of the pretrained model can be frozen (not updated during training) to retain learned features while training only the final layers for the new task.

Availability:

- Many pretrained models are available in popular deep learning frameworks (like TensorFlow and PyTorch), making it easy to implement them in various applications.

Advantages of Pre-trained Model:**Reduced Training Time:**

- Using a pretrained model significantly reduces the time and computational resources required for training a model from scratch, as the initial layers already capture useful features.

Improved Performance:

- Pretrained models often achieve better performance on smaller datasets because they leverage the knowledge gained from the larger dataset, they were trained on.

Ease of Use:

- They provide a convenient starting point for developers and researchers who may not have the resources to train large models themselves.

Generalization:

- Pretrained models tend to generalize better on new tasks due to their exposure to diverse data during initial training.

Example of Pretrained Models:

- Image Classification: Models like MobileNet, EfficientNetB0, VGGNet, ResNet, are commonly pretrained on ImageNet.
- Natural Language Processing: Models like BERT, GPT, and RoBERTa are pretrained on large text corpora and can be fine-tuned for various language tasks (e.g., sentiment analysis, text summarization).
- Object Detection: Models like Faster R-CNN and YOLO have pretrained weights.

2.8.1 Transfer learning

A machine learning technique called transfer learning makes use of information obtained while training one model and applies it to a separate but related job. It's a potent method that has gained popularity recently because of its capacity to improve model performance, shorten training times, and frequently use less labeled data. Here is a thorough justification of transfer learning: The idea that models might use the knowledge acquired from addressing one problem

to help solve another, related problem more successfully is the main driving force behind transfer learning. For each new job in classical machine learning, models are trained from the start, requiring a significant amount of labeled data and computer power.

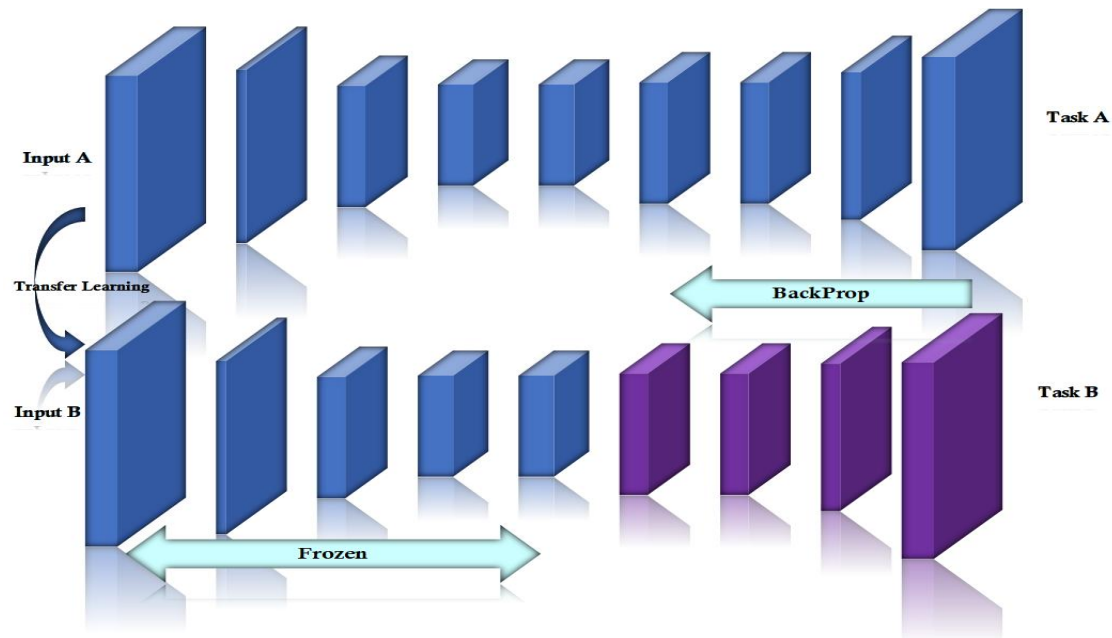


Figure 2.8.1: Transfer Learning

Two main steps are often involved in transfer learning:

Pre-training: A model learns to extract meaningful features or representations from the input data by being trained on a sizable dataset. A neural network, such as a convolutional neural network for images, or other machine learning models can be used as this pretrained model.

Fine-tuning: By further training the pretrained model on a smaller dataset associated with the new issue, the model is modified (fine-tuned) for the target job. Only the model 'stop layers' are changeable during this stage; the lower layers, which are in charge of feature extraction, are frequently kept frozen.

Based on the connection between the source and target tasks, transfer learning can be divided into many types:

Inductive Transfer Learning:

The most typical type of transfer learning is inductive, where the source and target tasks are distinct but the learned information is used to enhance the target task. Using a large image dataset for pretraining and fine-tuning for a particular image classification task, for instance.

Transductive Transfer Learning: In this scenario, the source and target tasks are identical, but the source domain information and target domain information could be different. For instance, utilizing insights from one camera's photos to enhance performance on photographs from another.

Unsupervised Transfer Learning: This occurs when the source and target tasks (such as clustering or dimensionality reduction) are both unsupervised.

Added advantages of transfer learning

Better Performance: When there is a lack of data, transfer learning frequently produces models that perform better than those that are trained from scratch

Less Labeled Data Needed: Compared to training from scratch, fine-tuning requires less labeled data.

Faster Training: Because the feature extraction layers of pre-trained models have already acquired relevant representations, training time is reduced.

Domain Adaptation: Transfer learning can customize models for various domains, strengthening their resistance to variations in data distribution. Resources are saved, including the processing power and knowledge necessary for large-scale model training.

The use of transfer learning: Transfer learning is frequently employed in various disciplines, including computer vision and natural language processing. Examples comprise:

- Using ResNet, VGG, or BERT deep neural networks that have been pretrained for a variety of applications.
- Language models for sentiment analysis, translation, and other uses are being fine-tuned.
- Using Transfer learning to diagnose diseases using medical imaging.
- Utilizing pre-trained models for fraud detection, recommendation engines, and sentiment analysis.

Challenges and Things to Think About

- Transfer learning is effective, but it can be difficult to choose the best pretrained model, specify the fine-tuning technique, and handle domain differences.
- The amount of adaptation versus reuse must be balanced. Avoid over-adapting to the source domain or job.
- When transferring knowledge from one domain to another, especially when working with sensitive material, ethical and privacy considerations are crucial.

Transfer learning involves using the knowledge gained and quick advancement from a source task to enhance learning and growth for a new target task. Utilizing the properties and traits of the source task, which will be applied and mapped onto the target work, is how knowledge is applied. Negative transfers, on the other hand, happen when the transfer mechanism causes the performance of the new target task to decline. Being able to give and ensure the positive transfer between activities that are connected while avoiding the negative transfer between tasks that are unrelated is one of the main issues when using transfer learning methods.

2.8.2 Fine-tuning

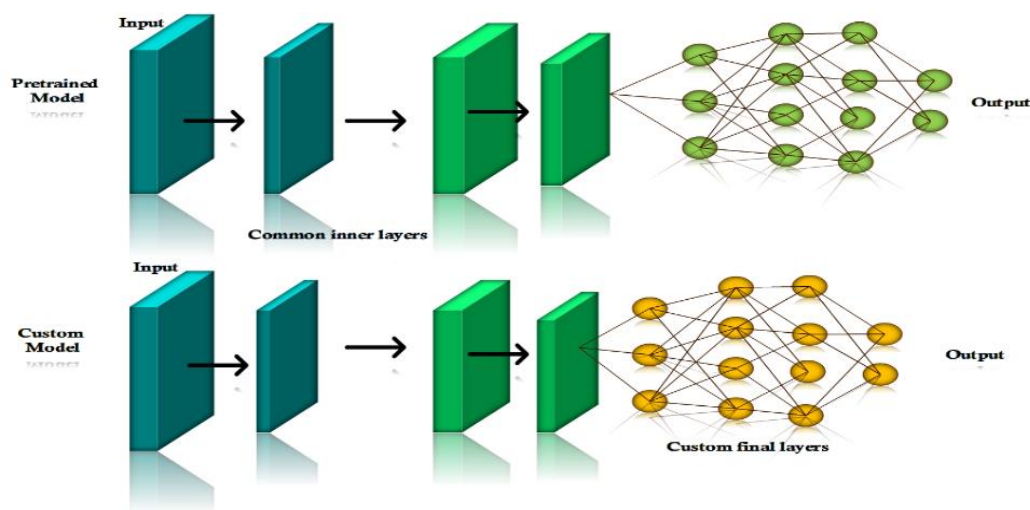


Figure: 2.8.2 Fine-tuning

Fine-tuning is a crucial technique in machine learning, particularly within the realm of transfer learning. It involves taking a pre-trained model—one that has already been trained on a large dataset—and adapting it to perform well on a new, often related, task or dataset. Fine-tuning leverages the knowledge the model has already acquired, allowing for more efficient training and often achieving better performance with less data and computational resources compared to training a model from scratch.

Steps to Fine-Tune a Pre-trained Model

Select a Pre-trained Model: Choose a model that has been trained on a large and relevant dataset. For example, use MobileNet for image classification tasks or BERT for NLP tasks.

Modify the Model Architecture: Remove the top layers of the pre-trained model that are specific to the original task. Add new layers suitable for the new task (e.g., a new classification layer with the appropriate number of output classes).

Freeze Base Layers: Initially, freeze the weights of the pre-trained layers to prevent them from being updated during the initial training phases.

Compile the Model: ConFigureure the model with appropriate loss functions, optimizers, and metrics.

Train the Model:

Initial Training: Train the newly added layers while keeping the pre-trained layers frozen.

Fine-Tuning: Gradually unfreeze some of the pre-trained layers and continue training with a lower learning rate to fine-tune the model's weights for the new task.

Evaluate and Optimize: Assess the model's performance on validation data, adjust hyperparameters as necessary, and employ techniques like early stopping to prevent overfitting.

2.8.3 Feature Extraction

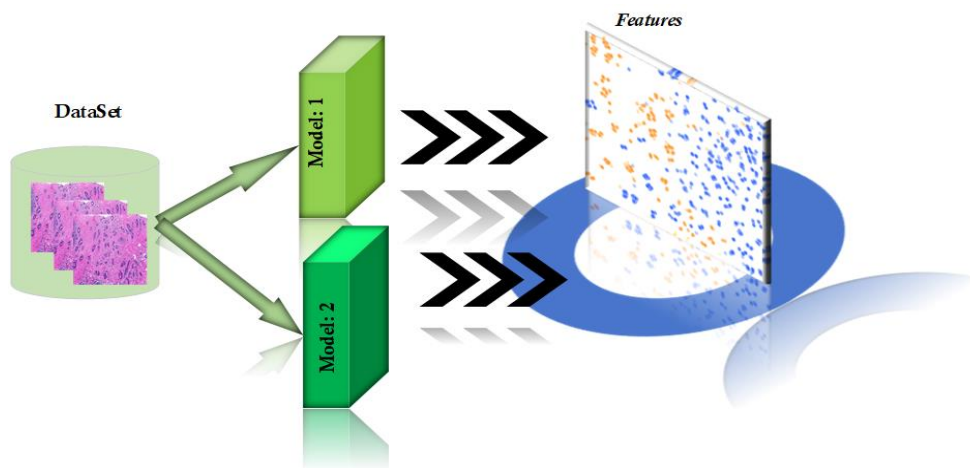


Figure 2.8.3 Feature Extraction Process

Feature extraction is a critical step in machine learning pipelines, particularly in image-based tasks such as breast cancer histopathology analysis. In this research, feature extraction is performed by leveraging deep learning models pre-trained on large image datasets. These models are used to capture complex, high-level visual patterns from histopathology slides, transforming raw pixel data into structured feature vectors. In this study, multiple pre-trained convolutional neural networks (CNNs) are used as feature extractors. These CNNs, trained on vast image datasets such as ImageNet, are capable of learning highly generalized visual features. The specific models employed in the ensemble are, MobileNet and EfficientNetB0 model. These CNNs are not fine-tuned but instead used in their pre-trained state, with weights frozen to ensure their general-purpose feature extraction capabilities are retained. Only the penultimate fully connected layers (or the output of the global average pooling layers, depending on the model architecture) are used to derive feature vectors from each histopathology image.

The feature extraction process can be summarized in the following steps:

Input Pre-processing: Each histopathology image is resized and normalized to match the input size requirements of the pre-trained networks. Standard normalization techniques, such as subtracting the dataset mean and dividing by the standard deviation, are applied.

Feature Vector Generation: For each image, feature vectors are extracted from the pre-trained CNNs. The output of the selected layer from each model serves as a high-dimensional feature representation. These vectors capture diverse characteristics, including texture, shape, and colour patterns, which are vital for distinguishing between malignant and benign tissue structures.

Dimensionality Reduction (if applicable): To reduce computational complexity and eliminate redundant information, a dimensionality reduction technique such as Principal Component Analysis (PCA) may be applied to the concatenated feature vectors. This step helps in retaining the most informative components while reducing noise.

Feature Concatenation: The feature vectors from each model are concatenated to form a comprehensive feature set for each image. This stacked feature set is used as input to the meta-learner in the ensemble framework. The combination of features from different pre-trained models ensures a diverse representation of the image data, contributing to the robustness and accuracy of the classifier.

2.9 Ensemble Learning

Ensemble learning combines multiple learners to improve predictive performance. It has been adopted in response to issues resulting from limited datasets.

Ensemble learning is a machine learning technique that aggregates two or more learners (e.g. regression models, neural networks) in order to produce better predictions. In other words, an ensemble model combines several individual models to produce more accurate predictions than a single model alone.[35] Ensemble learning rests on the principle that a collectivity of learners yields greater overall accuracy than an individual learner.[36]

A note on terminology: base learner, base model, and, in some cases, base estimator refers to the individual model or models used to in ensemble algorithms. Literature further divides base learners into strong learner and weak learners. weak models or learners are defined as those that perform little better than random guessing. For binary classification problems, weak

classifiers are more formally those that achieve approximately fifty percent accuracy. By contrast, strong models or learners achieve excellent predictive performance, which in binary classification is formalized as equal to or greater than eighty percent accuracy.[37]

Necessity of ensemble learning

Bias-variance tradeoff: Bias-variance tradeoff is a well-known problem in machine learning and a motivating principle behind many regularization techniques. I can define them as:

Bias: measures the average difference between predicted values and true values. As bias increases, a model predicts less accurately on a training dataset. High bias refers to high error in training. Optimization signifies attempts to reduce bias.

Variance: measures the difference between predictions across various realizations of a given model. As variance increases, a model predicts less accurately on unseen data. High variance refers to high error during testing and validation. Generalization refers to attempts to reduce variance. Bias and variance thus inversely represent model accuracy on training and test data respectively. They are two of three terms that comprise a model's total error rate, the third being irreducible error. This third term denotes error resulting from inherent randomness in a dataset. Total model error can be defined by the formula:

Many models versus one: Any one model training algorithm consists of numerous variables e.g., training data, hyperparameters, and so forth—that affect the consequent model's total error. Thus, even a single training algorithm can produce different models, each with their own bias, variance, and irreducible error rates. By combining several diverse models, ensemble algorithms can yield a weaker overall error rate while retaining each individual model's own complexities and advantages, such as notably low bias for a specific data subset. Research suggests that, in general, the greater diversity among combined models, the more accurate the resulting ensemble model. Ensemble learning can thus address regression problems such as overfitting without trading away model bias. Indeed, research suggests that ensembles comprised of diverse under-regularized models (i.e., models that overfit to their training data) outperform single regularized models. Moreover, ensemble learning techniques can help resolve issues stemming from high-dimensional data, and so effectively serve as an alternative to dimensionality reduction.

Types of ensemble models:

Literature widely categorizes ensemble learning methods in machine learning into two groups: parallel and sequential.

Parallel: methods train each base learner apart from the others of the others. Per its name, then, parallel ensembles train base learners in parallel and independent of one another.

Sequential: methods train a new base learner so that it minimizes errors made by the previous model trained in the preceding step. In other words, sequential methods construct base models sequentially in stages. Parallel methods are further divided into homogenous and heterogenous methods. Homogenous parallel ensembles use the same base learning algorithm to produce all of the component base learners. Heterogenous parallel ensembles use different algorithms to produce base learners.

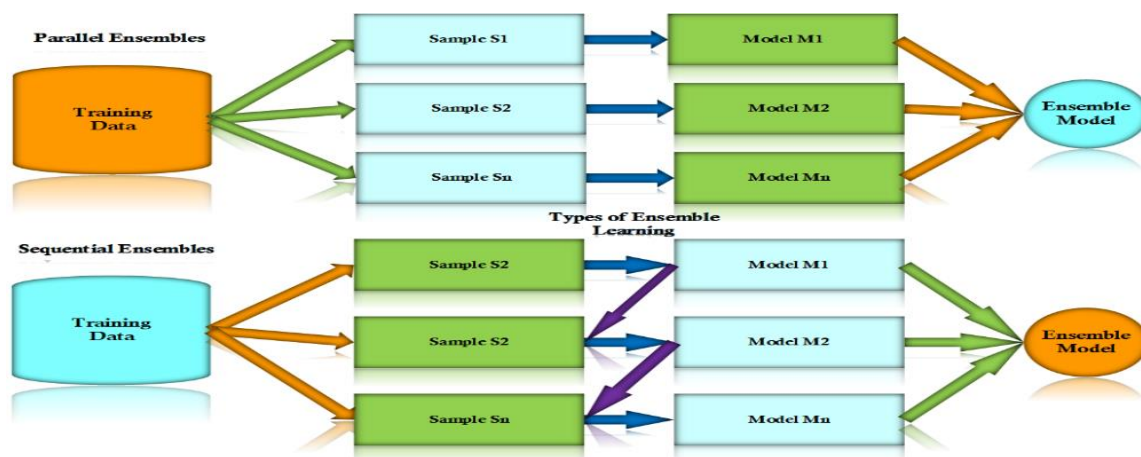


Figure 2.9: Types of Ensemble Learning

2.10 Ensemble Learning Techniques

Perhaps three of the most popular ensembles learning techniques are bagging, boosting, and stacking. In fact, these together exemplify distinctions between sequential, parallel, homogenous, and heterogenous types of ensemble methods.

Note that this overview is not exhaustive; there are several additional ensemble methods, such as blending and weighted average ensembles. This is merely meant to survey some of the more prominent methods in literature.

2.10.1 Stacking

Stacking, or stacked generalization, is a heterogeneous parallel method that exemplifies what is known as meta-learning. Meta-learning consists of training a meta-learner from the output of multiple base learners. Stacking specifically trains several base learners from the same dataset using a different training algorithm for each learner. Each base learner makes predictions on an unseen dataset. These first model predictions are then compiled and used to train a final model, being the meta-model.

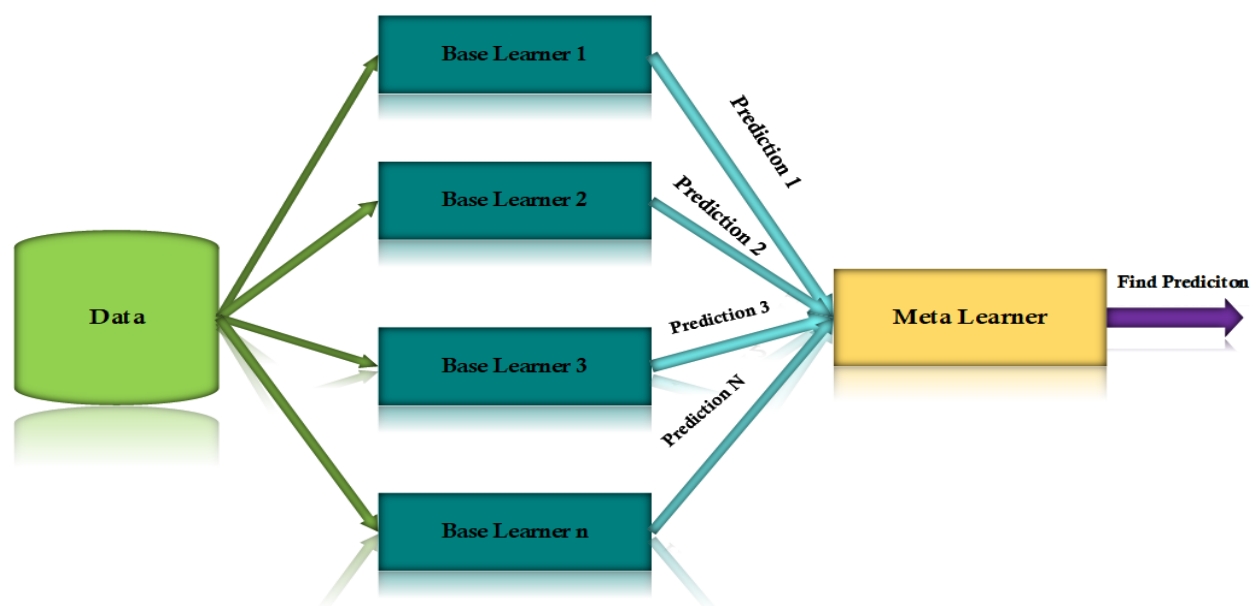


Figure 3.10.1: Stacking

Note the importance of using a different dataset from that used to train the base learners in order to train the meta-learner. Using the same dataset to train the base learners and the meta-learner can result in overfitting. This can require excluding data instances from the base learner training data to serve as its test set data, which in turn becomes training data for the meta-learner. Literature often recommends techniques such as cross-validation to ensure these datasets do not overlap. Much as bagging, the `sklearn.ensemble` module in Python provides various functions for implementing stacking techniques.

2.10.2 Bagging

Bagging is a homogenous parallel method sometimes called *bootstrap aggregating*. It uses modified replicates of a given training data set to train multiple base learners with the same training algorithm. Scikit-learns ensemble module in Python contains functions for implementing bagging, such as Bagging Classifier.

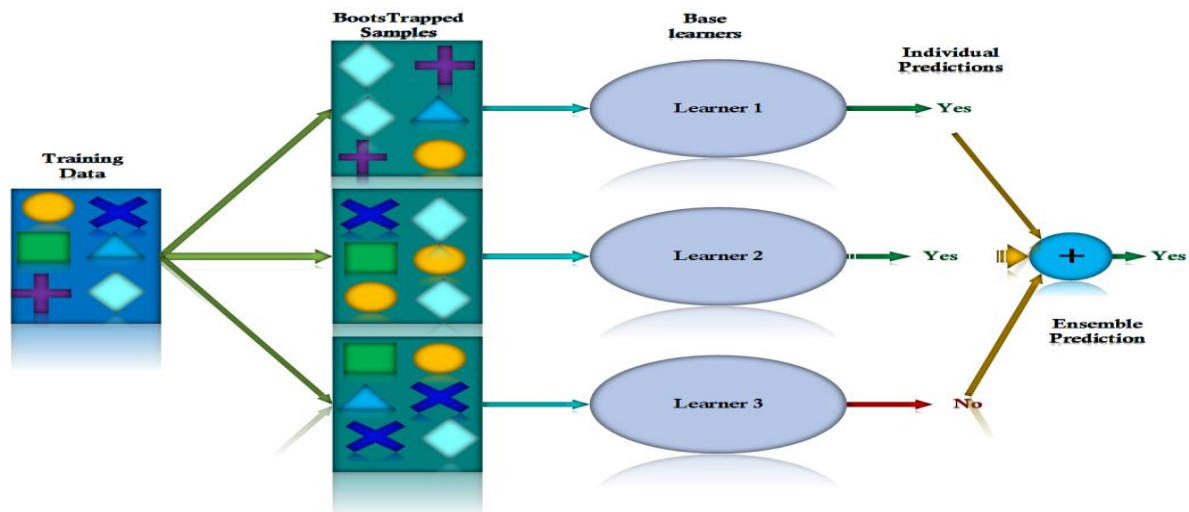


Figure 2.10.2: Bagging.

More specifically, bagging uses a technique called bootstrap resampling to derive multiple new datasets from one initial training dataset in order to train multiple base learners. How does this work? Say a training dataset contains n training examples. Bootstrap resampling copies n data instances from that set into a new subsample dataset, with some initial instances appearing more than once and others excluded entirely. These are bootstrap samples. Repeating this process x times produces x iterations of the original dataset, each containing n samples from the initial set. Each iteration of the initial set is then used to train a separate base learner with the same learning algorithm. Random forest is an extension of bagging that specifically denotes the use of bagging to construct ensembles of randomized decision trees. This differs from standard decision trees in that the latter samples every feature to identify the best for splitting. By contrast, random forests iteratively sample random subsets of features to create a decision node.

2.10.3 Boosting

Boosting algorithms are a sequential ensemble method. Boosting has many variations, but they all follow the same general procedure. Boosting trains a learner on some initial dataset, d . The resultant learner is typically weak, misclassifying many samples in the dataset. Much like bagging, boosting then samples instances from the initial dataset to create a new dataset (d_2). Unlike bagging, however, boosting prioritizes misclassified data instances from the first model or learner. A new learner is trained on this new dataset d_2 . Then a third dataset (d_3) is then compiled from d_1 and d_2 , prioritizes the second learner's misclassified samples and instances in which d_1 and d_2 disagree. The process repeats n times to produce n learners. Boosting then combines and weights all the learners together to produce final predictions. Boosting algorithms largely differ in how they prioritize erroneously predicted data instances when creating a new dataset.

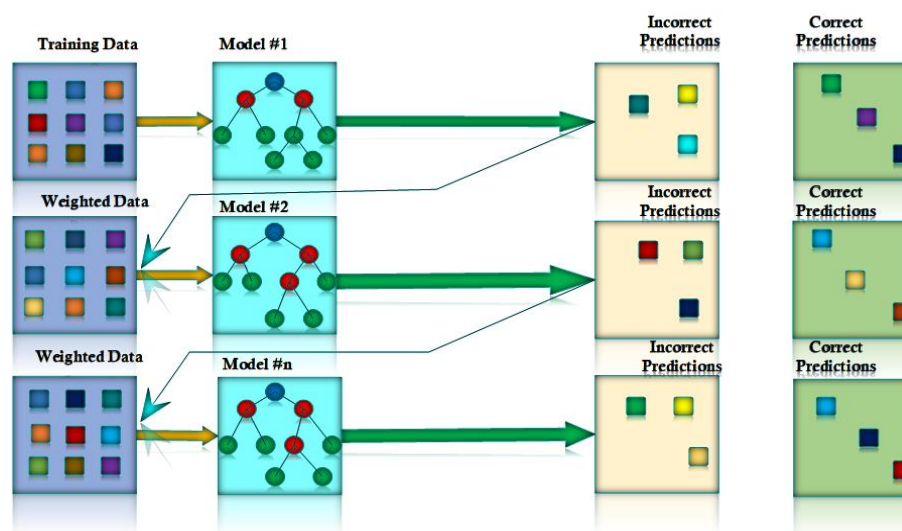


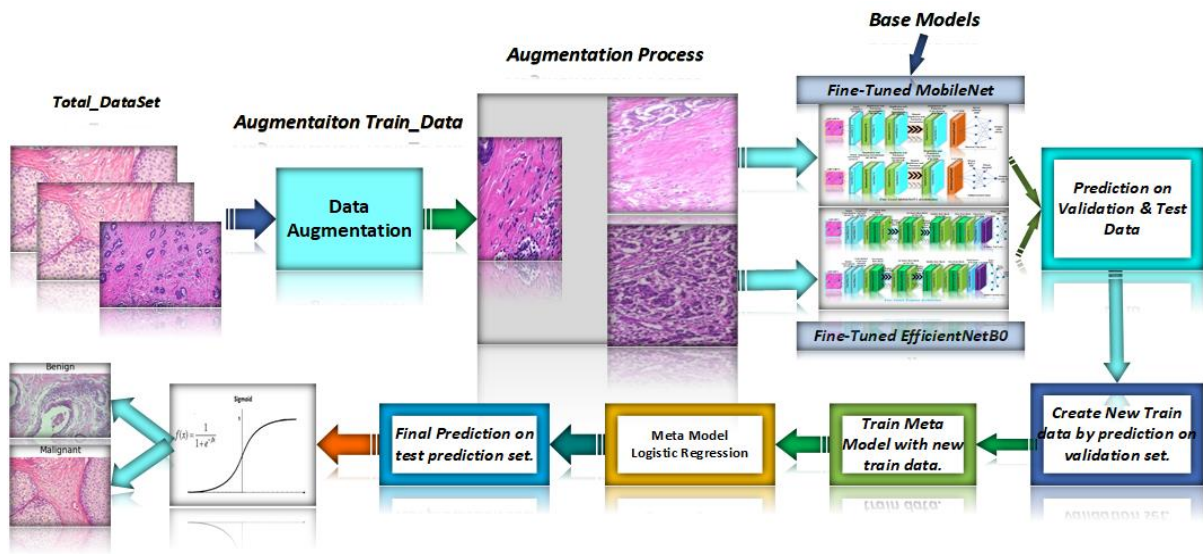
Figure 2.10.3: Boosting.

Two of the most prominent boosting methods may illustrate this Adaptive boosting (AdaBoost) weights model errors. That is, when creating a new iteration of a dataset for training the next learner, AdaBoost adds weights to the previous learner's misclassified samples, causing the next learner to prioritize those misclassified samples. Gradient boosting uses residual errors when training new learners. Rather than weight misclassified samples, gradient boosting uses residual errors from a previous model to set target predictions for the next model. In this way, it attempts to close the gap of error left by one model. Unfortunately, sklearn contains no pre-defined functions for implementing boosting. The Extreme Gradient Boosting (XGBoost) open-source library, however, provides code for implementing gradient boosting in Python.

Chapter 3: Methodology

In this chapter I have define the specific approaches, techniques, and processes used in this research.

3.1 Overview of Proposed Approach



Proposed stacking method architecture for classification of breast cancer using deep learning.

Figure 3.1: Proposed Approach

Proposed Model is an Ensemble model with Stacking technique. There are two stages in this entire model. In the first stage two pre-trained CNN models combined, these are MobileNetV1 and EfficientNetB0V1. These are called base model. And in meta model stage used Logistic Regression for final predictions.

Base Models: In the proposed model MobileNet and EfficientNetB0 model used. Where base models are train by train dataset and make prediction on validation data set for making train data set to meta learner, which make final prediction for test data set.

Meta Learner: Logistic Regression. Which used as Meta learner for final prediction in proposed model. This model train with a new tarin data set which is the combination of prediction from multiple base models. Finally meta learner makes prediction for test data set.

This section provides a high-level summary of the methodology developed in the thesis, outlining the novel approach taken to improve breast cancer histopathology analysis using ensemble learning.

The proposed approach leverages the pair of pre-trained deep learning models, which have demonstrated strong performance in various image classification tasks. Instead of relying on a single model, this research explores an ensemble learning technique, specifically a stacking method, to combine the strengths of multiple pre-trained models. The stacked ensemble consists of several Convolutional Neural Networks (CNNs), such as MobileNet, EfficientNetB0 and Inception, that have been pre-trained on large image datasets (e.g., ImageNet) and then fine-tuned on histopathological images of breast cancer.

The output predictions from these base models are combined using a logistic regression model, which serves as a meta-learner. In this stacking framework, each base model independently processes the input histopathological images and generates its prediction. These predictions are then fed into the logistic regression meta-learner, which learns to optimally combine the outputs of the base models and make the final classification decision (benign vs. malignant).

The rationale behind this approach is that each pre-trained model may capture different features and patterns in the data, and combining them through stacking can improve overall classification accuracy and robustness. Logistic regression, as a simple yet effective meta-learner, is used to High the contributions of each base model's predictions, ensuring that the final ensemble output benefits from the strengths of the individual models.

This ensemble strategy is evaluated on a breast cancer histopathology dataset to assess its performance in terms of standard evaluation metrics such as accuracy, precision, recall, and AUC-ROC. The approach aims to demonstrate that stacking multiple pre-trained deep learning models, with logistic regression as the meta-learner, can outperform individual models and enhance the reliability of automated breast cancer diagnosis.

3.2 Data Collection and Preprocessing

This This section provides a high-level summary total data collection to pre-processing.

3.2.1 Dataset description

The Breast Cancer Histopathological Image Classification (BreakHis) dataset is one of the most widely used public datasets for breast cancer research. It provides a diverse collection of breast cancer histopathological images, enabling researchers to develop, evaluate, and compare machine learning models for the automated classification of breast cancer tissue samples. As breast cancer remains one of the leading causes of cancer-related deaths among women globally, the BreakHis dataset plays a vital role in advancing diagnostic methods by providing a rich resource for training and testing artificial intelligence (AI) models.

This dataset is specifically designed for the binary classification of benign and malignant breast cancer tumors and is subdivided into various magnification levels to reflect the different zoom

levels used in histopathological analysis. In this section, I provide a comprehensive description of the BreakHis dataset, including its structure, composition, image characteristics, and the various applications it has been used for in breast cancer histography analysis.

3.2.2 Data overview

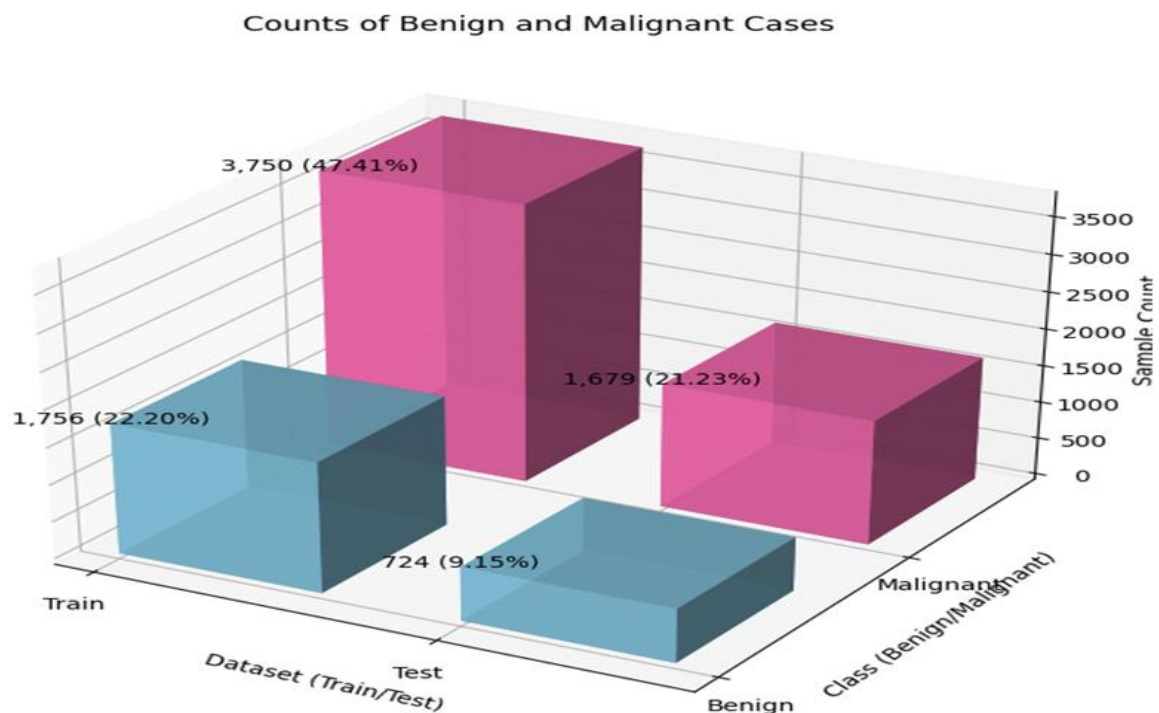


Figure: 3.2.2 Visual Data Overview

The BreakHis dataset was introduced by Spanhol et al to facilitate research in the area of breast cancer histopathology image analysis using machine learning techniques. The dataset contains a total of 7,909 images of breast tissue, divided into two main classes: benign and malignant tumors. The dataset is organized into four magnification levels—40x, 100x, 200x, and 400x—each of which reflects a different zoom level used by pathologists to examine tissue samples. This multi-scale representation of tissue samples is crucial for the development of automated systems that can detect cancer across various levels of detail.[30]

The images in the dataset are collected from breast tissue samples that are stained using Hematoxylin and Eosin (H&E), a widely used staining method in histopathology that highlights cellular structures and tissue morphology. The images are acquired using a standard optical microscope and captured at different magnification levels, allowing researchers to analyze tissue samples at varying levels of resolution.

3.2.3 Dataset Composition

The BreakHis dataset is organized into the following key components:

Number of Images: **7,909** breast tissue images

Number of Classes: **2 (Benign, Malignant)**

Number of Subclasses: 8 (**4 benign tumor types and 4 malignant tumor types**)

Magnification Levels: 4 (**40x, 100x, 200x, and 400x**)

Each image is assigned to one of the two main classes: benign or malignant. These two classes are further divided into eight subclasses, four for benign and four for malignant tumors:

Benign Tumors:

- Adenosis
- Fibroadenoma
- Tubular Adenoma
- Phyllodes Tumor

Malignant Tumors:

- Ductal Carcinoma
- Lobular Carcinoma
- Mucinous Carcinoma
- Papillary Carcinoma

Magnification Level	Benign Tumors	Malignant Tumors	Total Images
40x	625	1371	1996
100x	644	1437	2081
200x	623	1390	2013
400x	588	1231	1819
Total	2480	5429	7909

Figure: 3.2.3 Dataset Composition Table

Image Characteristics: Each image in the BreakHis dataset is a high-resolution histopathological image of a breast tissue sample stained with H&E, showcasing various cellular and tissue structures that are crucial for diagnosis. Key characteristics of the images are as follows:

Resolution: The images in the dataset are of varying resolutions, typically ranging from 700x460 to 2048x1536 pixels. This high resolution allows for the detailed examination of tissue structures.

Color_Information: The H&E staining results in distinct color patterns in the images. The nuclei of cells are stained dark purple due to the hematoxylin, while the cytoplasm and other extracellular components are stained pink by the eosin dye. These color differences provide important visual cues for pathologists and AI models to differentiate between benign and malignant tissues.

Texture and Morphology: The images exhibit a wide range of texture and morphological characteristics, which are critical for distinguishing between different tumor types. Benign tissues tend to have more uniform textures and ill-defined cellular structures, whereas malignant tissues often exhibit irregularities, higher cellular density, and structural disruptions.

Artifacts: As with any real-world medical imaging dataset, some images may contain artifacts, such as staining inconsistencies or areas of the image that are out of focus. These artifacts can present challenges for model training and evaluation, making the BreakHis dataset a realistic benchmark for developing robust AI models.

3.2.4 Magnification Levels:

One of the unique features of the BreakHis dataset is the inclusion of images at four different magnification levels (40x, 100x, 200x, and 400x). Each magnification level provides a different perspective on the breast tissue samples, allowing researchers to capture features at varying scales.

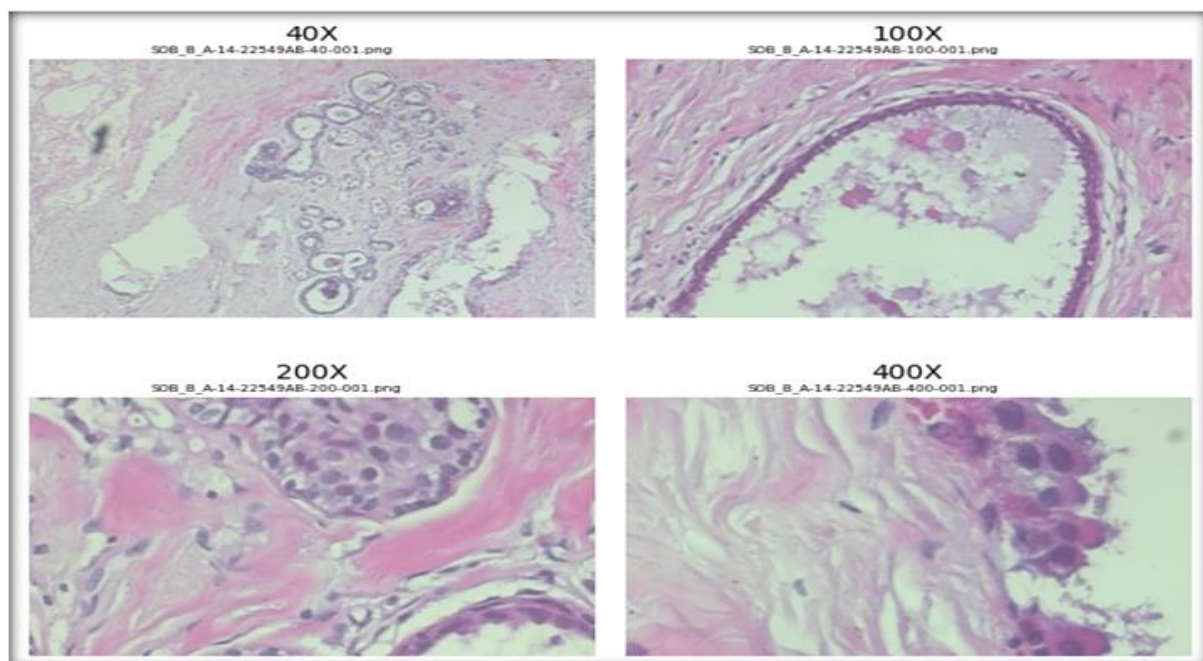


Figure 3.2.4: Magnification Levels

40x Magnification: The list magnification level, 40x, provides a broad view of the tissue architecture, allowing for the assessment of large-scale structural patterns such as ductal organization, tissue density, and overall cellular arrangement.

100x_Magnification: At 100x magnification, more detailed features become visible, such as the shape and arrangement of individual cells. This magnification level is particularly useful for assessing the overall cellularity and detecting larger-scale abnormalities in the tissue.

200x_Magnification: This intermediate magnification level allows for the examination of more detailed cellular structures, such as the morphology of the nuclei and cytoplasm. It is at this level that pathologists can begin to distinguish more subtle differences between benign and malignant tissues based on the appearance of individual cells and their relationships with neighboring cells.

400x_Magnification: The highest magnification level in the dataset, 400x, focuses on the fine-grained details of the tissue, such as nuclear size, shape, chromatin texture, and the presence of mitotic figures. This level is critical for identifying the microscopic features that are hallmarks of cancerous tissue, such as nuclear pleomorphism and high mitotic activity.

Each magnification level provides complementary information, and machine learning models that can integrate information across multiple scales are likely to perform better in classifying breast cancer tissues.

3.2.5 Data sample

The BreakHis dataset is a valuable resource for the development of machine learning and deep learning models for breast cancer histopathology analysis. Its rich collection of histopathological images at multiple magnification levels provides researchers with the opportunity to develop and evaluate a wide range of models, from traditional classifiers to state-of-the-art deep learning architectures. By providing a standardized dataset for breast cancer classification, BreakHis enables the comparison of different models and approaches, helping to advance the field of automated breast cancer diagnosis.

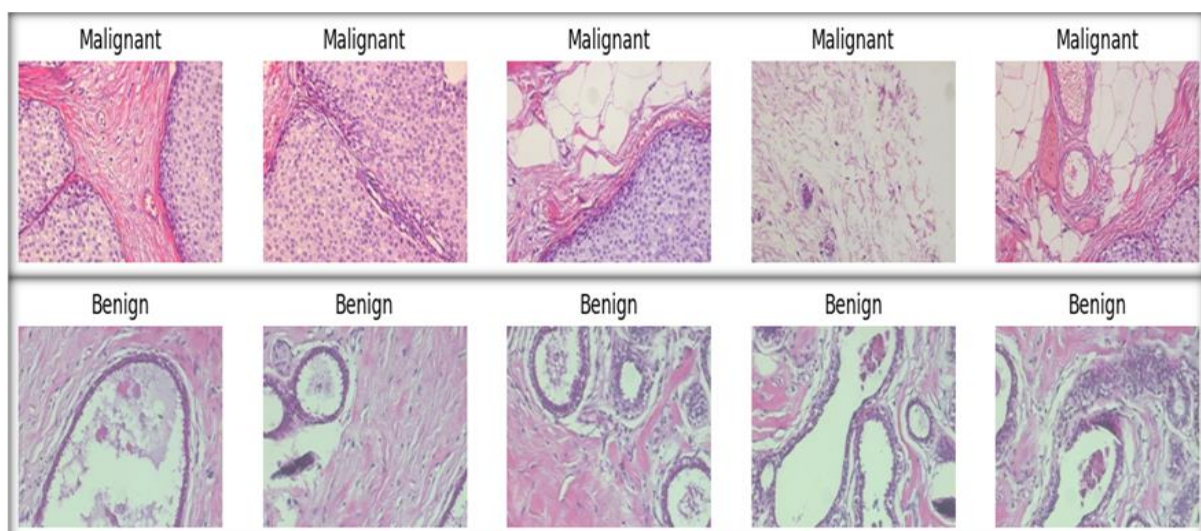


Figure 3.2.5: Data set Sample

While the dataset presents several challenges, such as class imbalance and limited size, these issues can be addressed through techniques like transfer learning, data augmentation, and ensemble learning. The insights gained from working with the BreakHis dataset have the

potential to improve breast cancer diagnosis and ultimately contribute to better patient outcomes.

3.2.6 Data Set Acquisition Process

There are various implementation details that must be taken into consideration in order to use the Breast Cancer Histopathological Image Classification. (BreakHis) dataset efficiently for research or machine learning activities. The BreakHis dataset can be properly prepared and used for machine learning or research on the histopathological image categorization of breast cancer by adhering to these implementation specifications.

Data Collection: Download the BreakHis dataset from a reputable repository or directly from the publisher's Inside (<http://www.prevencaoediagnose.com.br>). And make sure you have 9,109 PNG files representing the tissue from breast tumors. The advancement of machine learning and computer vision may provide more accurate categorization techniques for the histological evaluation of hematoxylin and eosin-stained sections. These techniques have great classification rates for automatically categorizing breast tissues [38]

Data Storage: Dataset which combination of CSV file and Image file, stored on my google drive and mount the drive file when I worked on google Colab. Depending on your processing needs and capabilities, store the dataset on your computer locally or in a cloud-based storage system.

Data Loading: To read and load photos from the dataset into your analytic or machine learning environment, create data loading functions or scripts. Then manage picture loading, you can utilize libraries for image processing such as OpenCV or PIL.

Data Balancing: In order to balance the classes in the training set, check for class imbalance and, if necessary, use strategies like oversampling, under sampling, or synthetic data generation.

Batching and data loading: During model training and evaluation, create data loading pipelines that effectively load and preprocess batches of photos. Use these pipelines to implement data transformation and augmentation.

Data documentation and versioning: in order to track alterations or updates over time, keep version control for the dataset. Make a report that details the dataset, its source, the preprocessing procedures, and any alterations that I made.

Data Security and Privacy: Make sure that the dataset is stored securely and that only people with permission are allowed access. Handle the dataset in accordance with privacy laws and industry best practices if it contains sensitive patient data.

Exploration and visualization of data: Learn more about the distribution of benign and cancerous samples, magnification factors, and other pertinent aspects by exploring the collection through visualizations.

Data Logging and Monitoring: Implement logging and monitoring to track data access, changes, and any issues that may arise during data processing.

Data quality control: Implement data quality tests to find and correct any anomalies, mistakes, or outliers in the data. Optimization of data storage, if applicable: Consider data storage optimization approaches to cut storage costs, depending on the size of the dataset.

Backup and recovery of data: Create regular backups of the dataset to guard against data loss due to hardware malfunctions or other unforeseen circumstances.

3.2.7 Data preprocessing

The classification of histological images was done by processing many patches of a specified size. Histological characteristics that can be seen up close separate cancer cells. The cytoplasm may also exhibit atypia, and the nucleus is frequently big and atypical. Additionally, there are obvious structural variations between healthy and sick tissues [39]. Here is an example of data preprocessing for the BreakHis dataset:

Data collection: The BreakHis dataset includes 9,109 microscopic pictures of breast tumor tissue that Ire gathered from 82 patients, as was previously mentioned. The photos in the data, which Ire taken at various magnifications (40X, 100X, 200X, and 400X), are kept in PNG format.

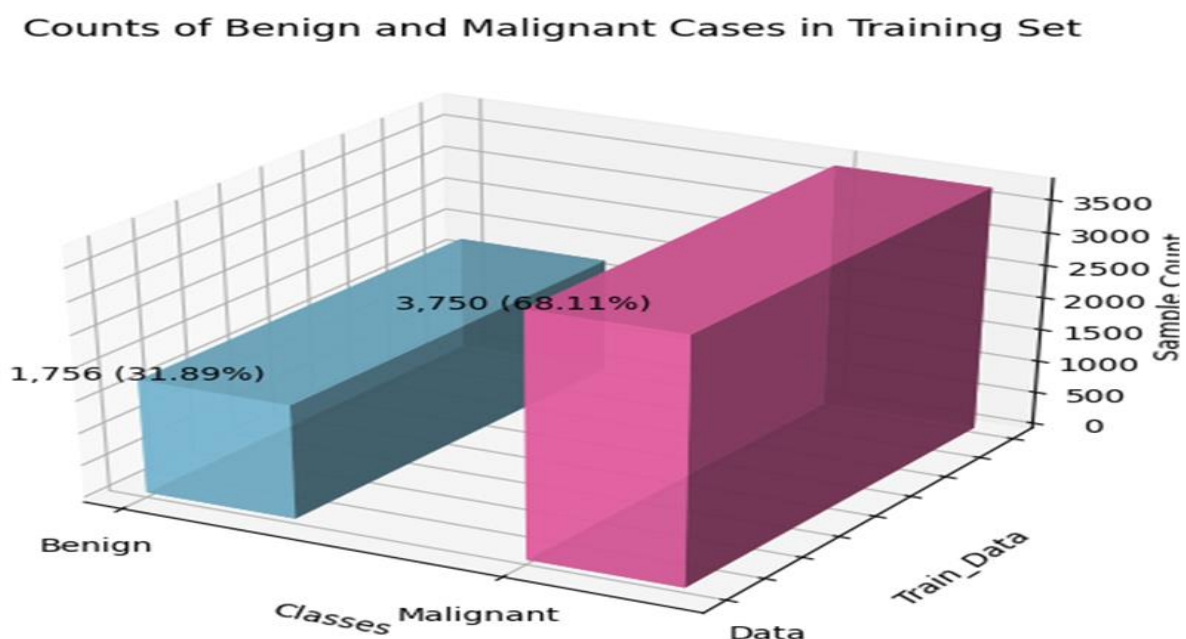


Figure 3.2.7 Train Data Set

It is crucial to inspect the dataset to comprehend its structure, features, and any potential problems before beginning any preprocessing. This entails: examining the PNG file format. confirming the photos' (700 x 460 pixel) dimensions.3. examining the bit depth (8 bits per channel). The are two set of data these are train and test show in Figurer 3.2.2 but from these two set I picked train set for total research.

Data balancing: If the classes are significantly out of balance (for example, there are more malignant than benign examples), you might want to use methods like oversampling, under

sampling, or creating synthetic samples. Keep a log of all preprocessing decisions and actions, as well as information about image scaling, normalization, and any data augmentation that was used. Depending on the application, you may use data augmentation techniques to fictitiously expand the size of your training dataset. Rotation, flipping, and zooming are common image enhancements. In train data set 31.89% benign and 68.11% malignant data. So, need to balancing data set. That's why I was selected some benign data from train set to train set.

Counts of Benign and Malignant Cases in Training Set

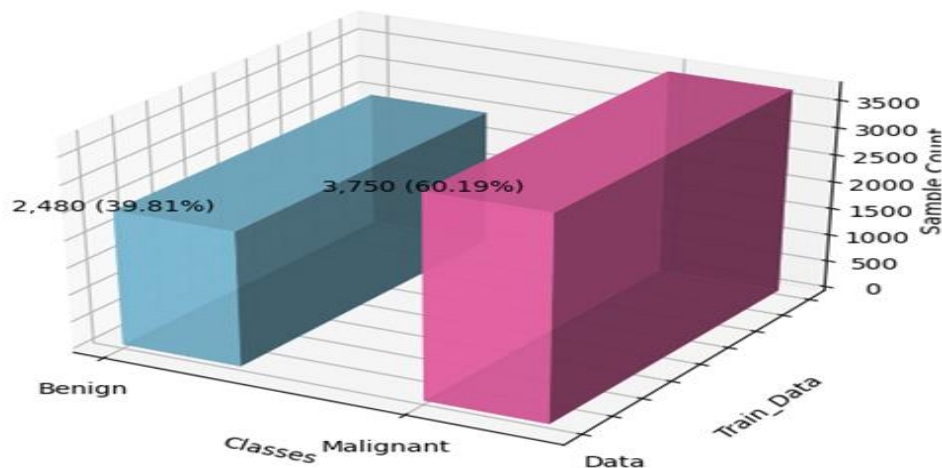


Figure 3.2.7: Train Data Balancing

Data Cleaning: For the BreakHis dataset, the following actions may be taken: Taking Care of Missing Photographs: Verify that every one of the 9,109 photographs is there and that none are missing. Then Check the photos for any anomalies, corruption, or artifacts that could taint the analysis as part of the quality control process. Visual inspection or automated quality checks may be used in this.

3D Visualization of Image Issues

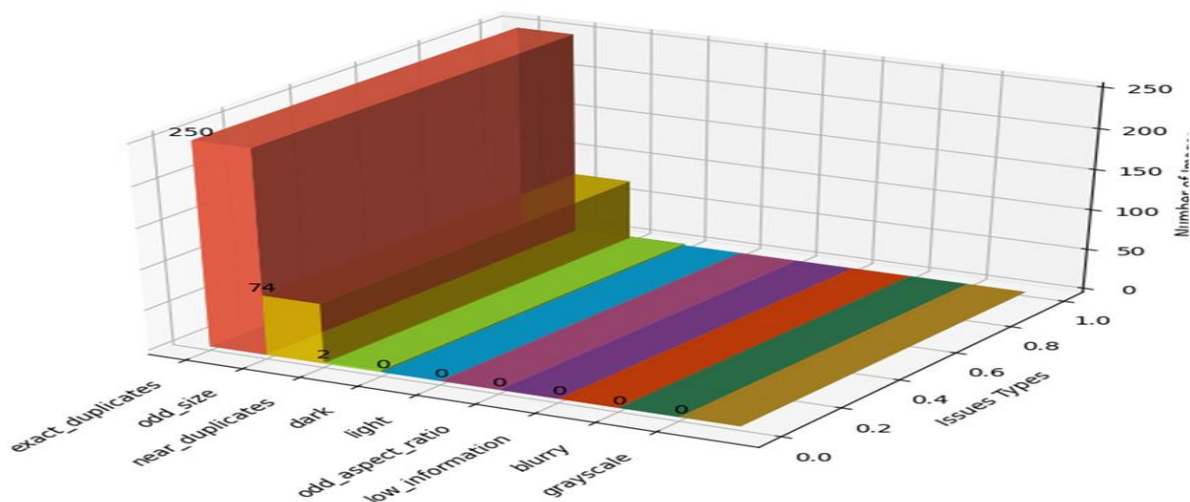


Figure 3.2.7: Checking Issues

Removing Unwanted Photographs: Any photographs that are deemed unnecessary or of poor quality may be taken out of the dataset. **Data Transformation:** You might need to do the following to get the photos ready for machine learning:

Image Issues Report	
issue_type	num_images
exact_duplicates	0
odd_size	0
near_duplicates	0
dark	0
light	0
odd_aspect_ratio	0
low_information	0
blurry	0
grayscale	0

Figure 3.2.7: Report After Clean

Counts of Benign and Malignant Cases in Training Set

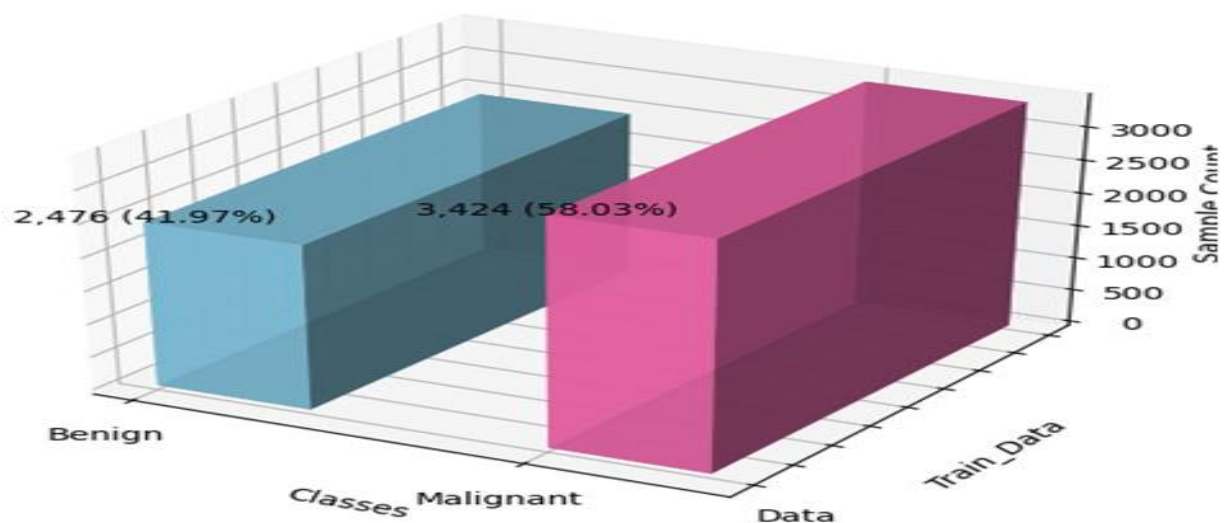


Figure 3.2.7: Train Data Set After Clean

Normalization: To guarantee uniformity in data representation, normalize pixel values, typically to a range between 0 and 1 for each channel. Encode labels onto the photos according to whether they show benign or malignant tumors. For tasks involving supervised learning, these labels are necessary. Splitting the data into sets for training, validation, and testing. Usually, a sizeable chunk is set aside for training (for example 70%), a smaller piece for

validation (for example, 20%) to allow for hyperparameter tuning, and a third portion (for example, 10%) to test the model and assess its performance.

Data Augmentation: I might use data augmentation techniques to fictitiously expand your training dataset, depending on the application. Rotation, flipping, and zooming are common image enhancements.

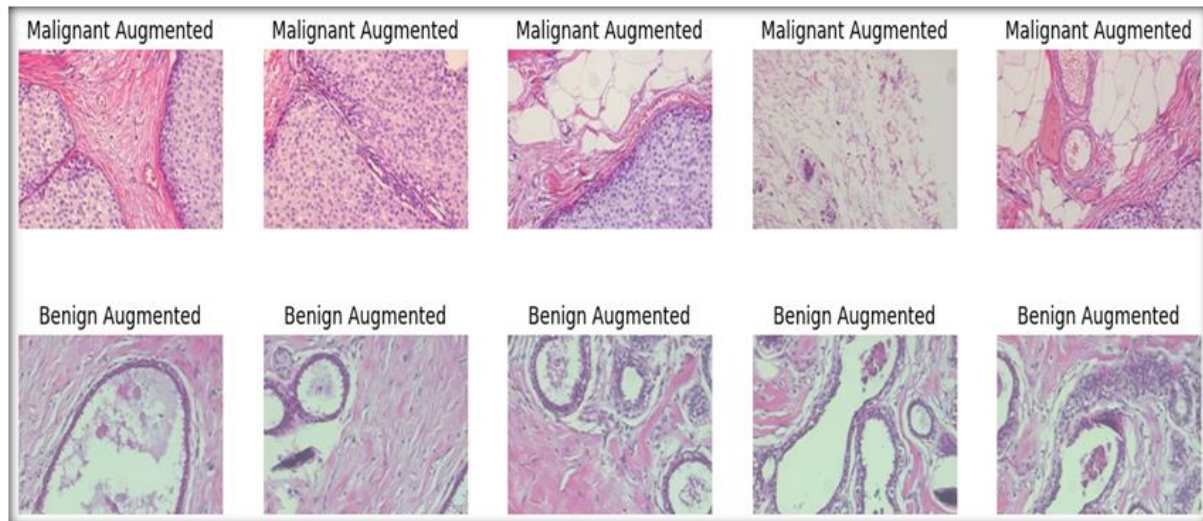


Figure 3.2.7: Augmented Data

Data Splitting: Create training, validation, and testing sets from the dataset. 70% for training, 20% for validation, and 10% for testing are typical splits to prevent class imbalance issues, make sure the splits retain a comparable distribution of benign and malignant samples.

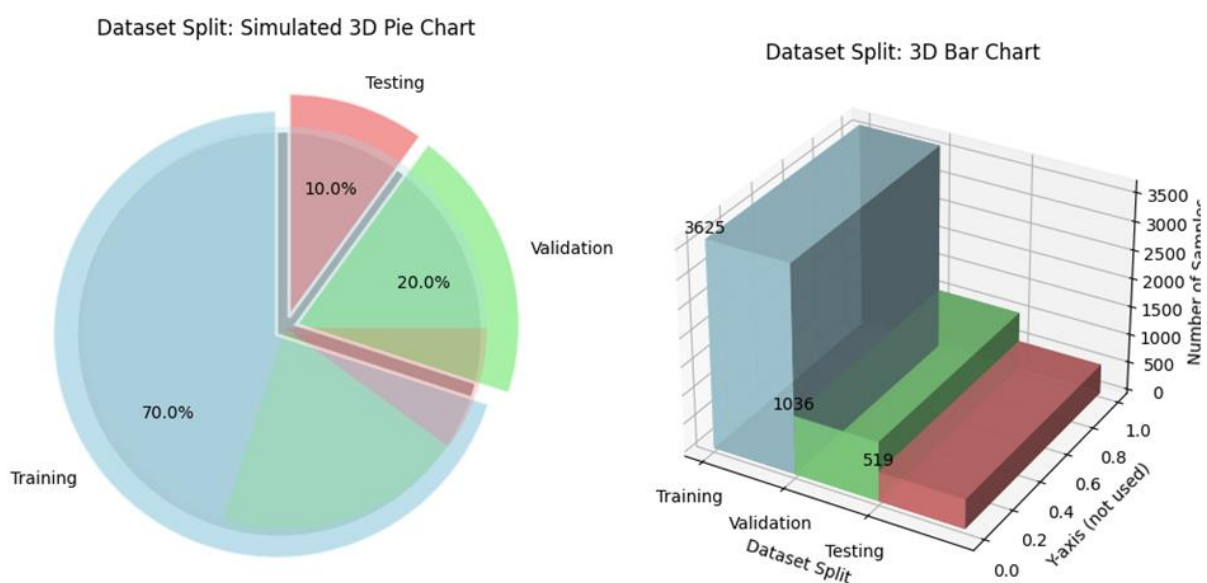


Figure 3.2.7: Data Splitting

Exploratory Data Analysis (EDA): Use EDA to learn more about the dataset by exploring relationships, visualizing distributions, and looking for any trends or potential 32 problems. In order to effectively use the BreakHis dataset for breast cancer histopathology image classification tasks, data preprocessing is a crucial step. It guarantees that the data is in the proper format and quality to be used for developing artificial intelligence models or performing more research, ultimately resulting in accurate and valuable findings.

3.3 Base Models: Pre-Trained Deep Learning Networks

This section describes the deep learning models used as the base models in the proposed ensemble framework. These models are pre-trained Convolutional Neural Networks (CNNs) that have been extensively trained on large image datasets like ImageNet, making them effective at feature extraction from images, even in medical domains like breast cancer histopathology. The section covers the selection, architecture, and fine-tuning of these models for the specific task of classifying histopathological images.

3.3.1 Selection of Pre-Trained Models

The selection of base models is critical to the success of the ensemble. In this thesis, several well-established CNN architectures, such as MobileNet and EfficientNetB0, are chosen as the base

models for the following reasons:

EfficientNetB0 Model:

EfficientNetB0 is the base model of the EfficientNet family, introduced by Mingxing Tan and Quoc V. Le in 2019. It utilizes a novel approach called *compound scaling*, which uniformly scales up the depth, width, and resolution of the network, achieving state-of-the-art results on various image classification tasks while being computationally efficient. EfficientNet models have been widely adopted for their balance between accuracy and efficiency.

Key Features of EfficientNetB0:

- **Compound Scaling:** EfficientNet scales the network in three dimensions—depth (number of layers), width (number of channels), and resolution (image size)—in a balanced way.
- **Mobile Inverted Bottleneck Convolution (MBConv):** A key building block of EfficientNet, the MBConv combines depthwise separable convolutions with residual connections, reducing the number of parameters while maintaining performance.
- **Swish Activation:** EfficientNet uses the Swish activation function, which helps to smooth the gradient flow and improve the learning process.

Application of EfficientNetB0 in Breast Cancer Histopathology:

Rationale for Using EfficientNetB0:

- **Efficiency:** EfficientNetB0 achieves high accuracy with a small number of parameters and lower computational costs, making it well-suited for processing high-resolution histopathology images.
- **Transfer Learning:** Pre-trained on ImageNet, EfficientNetB0 can be fine-tuned to extract relevant features for breast cancer histopathology images, such as tumor cell characteristics.
- **Adaptability to Large Datasets:** With its balanced scaling, EfficientNetB0 can handle large histopathology datasets, optimizing both the feature extraction process and the classification task.

MobileNet Model:

Overview: MobileNet is a family of light weight deep learning models designed for mobile and edge devices, known for their efficiency and performance in image classification tasks. Introduced by Google, MobileNet uses depth wise separable convolutions to significantly reduce the number of parameters and computation required while maintaining competitive accuracy. This characteristic makes it particularly suitable for applications where computational resources are limited, such as real-time mobile applications and embedded systems.

Key Features:

Depthwise Separable Convolutions: MobileNet employs depthwise separable convolutions, which break the convolution operation into two layers:

- **Depthwise Convolution:** Applies a single filter per input channel, resulting in lower computation.
- **Pointwise Convolution:** Follows the depthwise layer to combine the output channels using a standard 1×1 convolution.

This separation reduces the overall computational cost compared to traditional convolutional networks.

- **Light weight Architecture:** MobileNet's architecture is optimized for performance, providing a smaller model size and faster inference times. It achieves this by using fewer parameters and reducing the number of computations, making it well-suited for mobile devices.
- **Adjustable Width and Resolution:** MobileNet allows for tuning the model size and resolution through two hyperparameters:

- **Width Multiplier (α):** Reduces the number of channels in each layer, allowing for a trade-off between model size and accuracy.
- **Resolution Multiplier (ρ):** Adjusts the input image resolution, which can further reduce computational load while affecting accuracy.
- **Pre-Trained Lights:** MobileNet models are often pre-trained on large datasets like ImageNet, allowing them to leverage transfer learning. This is particularly useful for tasks with limited labelled data, such as breast cancer histopathology analysis, as it can help the model achieve good performance with less training data.

Application in Breast Cancer Histopathology: In your thesis, MobileNet can serve as a base model for the ensemble approach to analyse breast cancer histopathology images. The strengths of MobileNet, such as its efficiency and accuracy, make it suitable for:

- **Feature Extraction:** MobileNet can be used to extract relevant features from histopathology images, which can then be fed into a logistic regression meta-learner.
- **Ensemble Learning:** As part of an ensemble of pre-trained models, MobileNet can contribute to the overall predictive pair, enhancing the classification performance when combined with other models.

3.3.2 Architecture and Fine-Tuning

Once the base models are selected, they need to be fine-tuned for the task of breast cancer histopathology analysis. The pre-trained models, which have learned general image features from large datasets, are adapted to the specific histopathological images in this study. This process typically involves:

Fine-Tuning:

- **Replacing the Final Layer:** Since the original pre-trained models are designed to classify a large number of classes (e.g., 1000 classes in ImageNet), the final classification layer is replaced with a new output layer specific to this task. In this case, a binary classification layer (benign vs. malignant) is used.
- **Transfer Learning:** To leverage the pre-trained weights, transfer learning is applied by freezing the initial layers of the network, which have learned to extract general features like edges, textures, and shapes. Only the last few layers of the network are fine-tuned to adapt to the new task and learn more specific patterns related to breast cancer histopathology.
- **Optimization and Loss Functions:** The models are trained using appropriate optimization techniques, such as Adam or stochastic gradient descent (SGD), and a suitable loss function like binary cross-entropy, given the binary classification problem. The learning rates and batch sizes are tuned to achieve optimal performance.
- **Data Augmentation:** To prevent overfitting and improve the generalizability of the models, data augmentation techniques (such as rotation, flipping, scaling, and colour adjustments) are applied during training. This ensures that the models are exposed to a variety of transformations, which helps them better generalize to unseen data.

Parameter Summary EfficientNet B0 Model After Fine-Tuning:

Model: "sequential_2"

Layer (type)	Output Shape	Param #
efficientnetb0 (Functional)	(None, 4, 4, 1280)	4,049,571
global_average_pooling2d_2 (GlobalAveragePooling2D)	(None, 1280)	0
dense_4 (Dense)	(None, 256)	327,936
dense_5 (Dense)	(None, 1)	257

Total params: 13,049,248 (49.78 MB)
Trainable params: 4,335,741 (16.54 MB)
Non-trainable params: 42,023 (164.16 KB)
Optimizer params: 8,671,484 (33.08 MB)

Figure 3.3.2 Summary EfficientNetB0 Model

Parameter Summary of MobileNet Model After Fine-Tuning:

Model: "sequential_1"

Layer (type)	Output Shape	Param #
mobilenet_1.00_128 (Functional)	(None, 4, 4, 1024)	3,228,864
global_average_pooling2d_1 (GlobalAveragePooling2D)	(None, 1024)	0
dense_2 (Dense)	(None, 256)	262,400
dense_3 (Dense)	(None, 1)	257

Total params: 10,430,789 (39.79 MB)
Trainable params: 3,469,633 (13.24 MB)
Non-trainable params: 21,888 (85.50 KB)
Optimizer params: 6,939,268 (26.47 MB)

Figure 3.3.2 Summary of MobileNet Model

EfficientNetB0 Architecture:

EfficientNetB0's architecture can be divided into the following stages:

Stem Layer:

- The input image is passed through the first convolutional layer with a large kernel size (3x3) and a stride of 2 for initial downsampling, reducing the spatial dimensions while increasing the number of channels.

Convolutional Blocks (MBConv):

- The backbone of EfficientNetB0 is composed of several *Mobile Inverted Bottleneck* (MBConv) blocks, each of which applies depthwise separable convolutions.
- These blocks are optimized for efficiency, reducing the number of parameters while preserving performance. The depthwise convolutions handle spatial relationships, while pointwise convolutions manage feature learning.
- Each MBConv block includes residual connections, which improve gradient flow during training, aiding in faster convergence and deeper network performance.

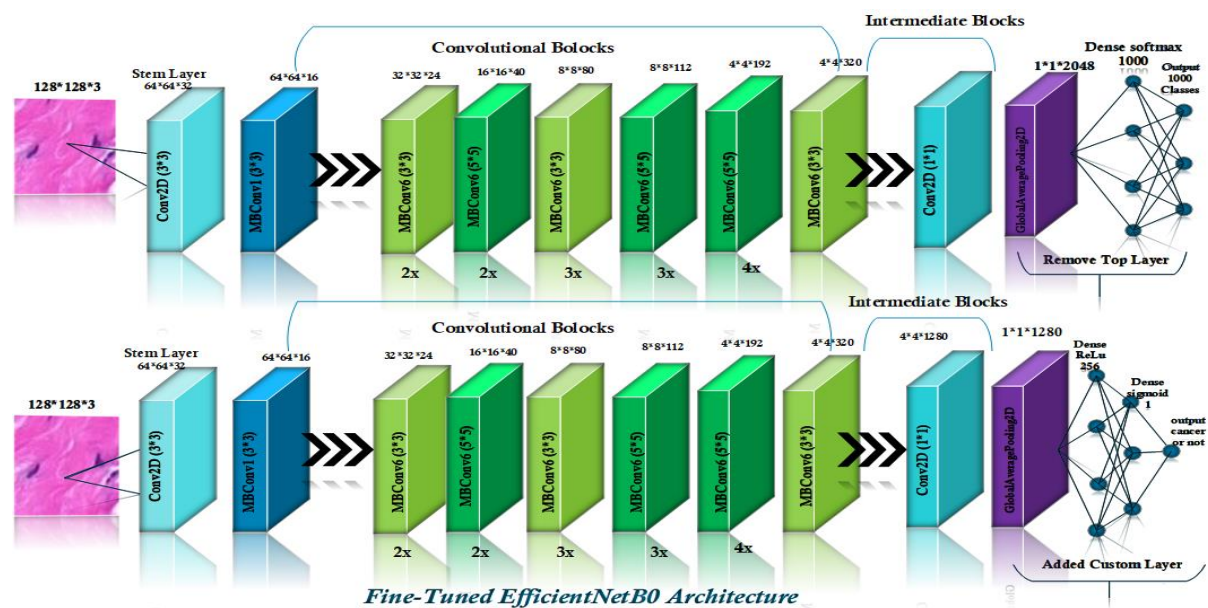


Figure: 3.3.2 Fine-Tuned EfficientNetB0 model Architecture

Intermediate Blocks:

- These blocks use a combination of expanded filters and shrinking channels, ensuring that complex features are extracted while maintaining computational efficiency. The compound scaling technique ensures that the depth, width, and input resolution are appropriately adjusted based on task requirements.

Global Average Pooling:

- After the final set of convolutional layers, the spatial dimensions are reduced using global average pooling, transforming the spatial feature maps into a single vector for classification.

Fully Connected Layer:

- The pooled features are passed through a fully connected (dense) layer, which maps the learned features to the target classes (in this case, benign or malignant tumors for breast cancer histopathology).

Softmax Layer:

- A softmax function is applied to the output of the dense layer to produce class probabilities, helping the model make final classification decisions.

Architecture of MobileNet-based Ensemble Model:

MobileNet Architecture Overview with fine-tuned:

Depth wise Convolution: This applies a single filter to each input channel, processing spatial information separately for each feature map. Unlike traditional convolutions, depth wise convolution doesn't mix information across channels.

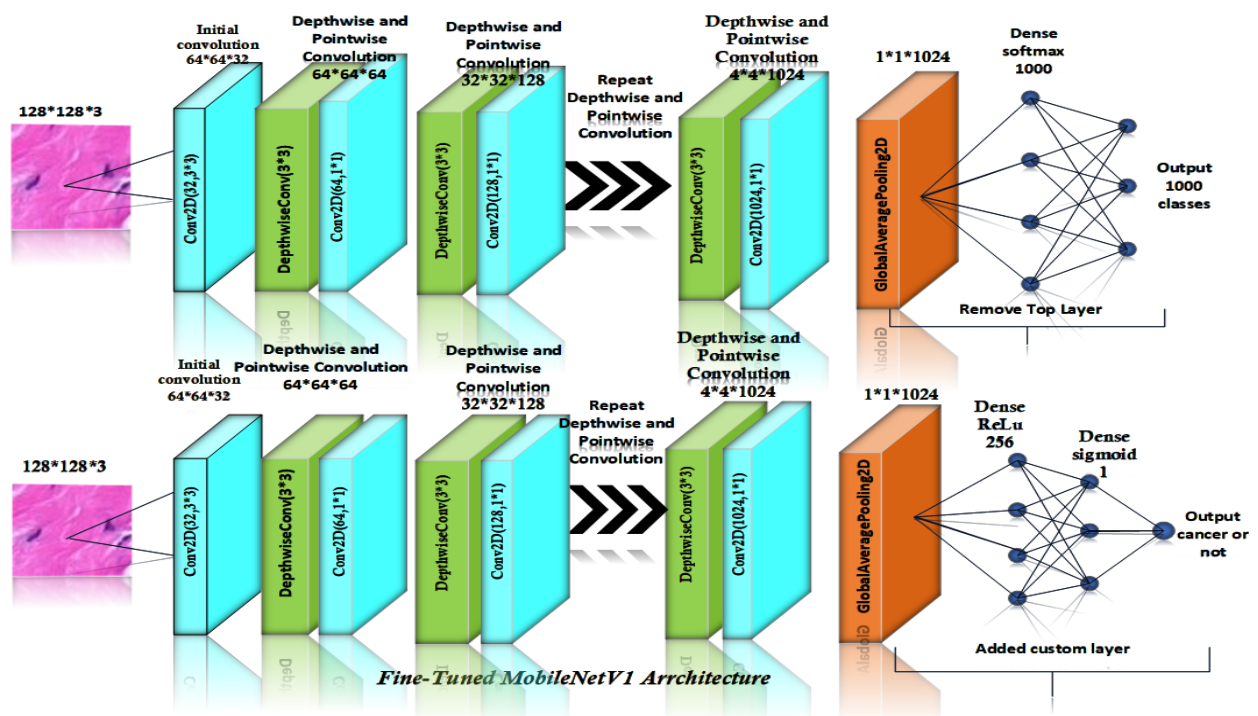


Figure: 3.3.2: MobileNet Model Architecture

Pointwise Convolution: A 1x1 convolution is then applied to combine the output of the depth wise convolution across different channels. This step captures cross-channel information and reduces the number of parameters needed for computation.

By breaking down standard convolution operations, MobileNet achieves an order of magnitude fewer parameters and floating-point operations per second (FLOPs) than traditional CNN architectures, such as VGG16 or ResNet, without significantly sacrificing accuracy.

The ensemble learning strategy combines multiple pre-trained deep learning models to enhance classification performance. By aggregating the predictions of several models, ensemble methods can reduce model variance, minimize overfitting, and improve generalization, especially on challenging datasets like breast cancer histopathology.

The ensemble model typically operates in the following manner:

Feature Extraction: MobileNet and other CNNs are employed as feature extractors, where each model processes the input histopathological images to extract relevant features.

Aggregation of Features: The features from MobileNet and other pre-trained models are aggregated, either by concatenation or averaging. This step allows the ensemble to leverage the unique strengths of different models, resulting in richer and more diverse feature

representations. Classification Layer: The aggregated features are passed through a fully connected layer and a SoftMax classifier to output the final prediction. The ensemble output can be computed via majority voting, weighted averaging, or a more sophisticated technique like stacking, where a meta-learner combines the predictions of individual models.

Performance Improvements through Ensemble Learning:

Studies show that ensemble learning, when applied to breast cancer histopathological images, can significantly outperform single models in terms of accuracy, sensitivity, and specificity. In particular, MobileNet's efficiency allows for the inclusion of multiple models in the ensemble without exceeding computational limits, ensuring that the system remains viable for clinical use. An ensemble model combining MobileNet, ResNet, and Inception has been shown to achieve higher diagnostic accuracy compared to any single model. In one study, an ensemble approach improved classification accuracy on the BreakHis dataset, a common breast cancer histopathological dataset, from 85% (single model) to 92% (ensemble).

MobileNet is an effective model for breast cancer histography analysis, particularly when used as part of an ensemble approach. Its light weight architecture and efficiency in feature extraction make it a valuable component in multi-model systems that aim to improve diagnostic accuracy. By combining MobileNet with other CNNs in an ensemble, it is possible to leverage the strengths of each model while maintaining computational feasibility, making it well-suited for real-time clinical applications.

3.4 Ensemble Learning Framework:

Process of Stacking: Stacking, also known as "Stacked Generalization," is a machine learning ensemble strategy that integrates many models to improve the model's overall performance. The primary idea of stacking is to feed the predictions of numerous base models into a higher-level model known as the meta-model or blender, which then combines them to get the final forecast.

3.4.1 Stacking Methodology:

Here's a detailed description of how stacking works:

Preparing the Data: The first step is to prepare the data for modeling. This entails identifying the relevant features, cleaning the data, and dividing it into training and validation sets.

Model Selection: The following step is to choose the base models that will be used in the stacking ensemble. A broad selection of models is typically chosen to guarantee that they produce different types of errors and complement one another.

Training the Base Models: After selecting the base models, they are trained on the training set. To ensure diversity, each model is trained using a different algorithm or set of hyperparameters.

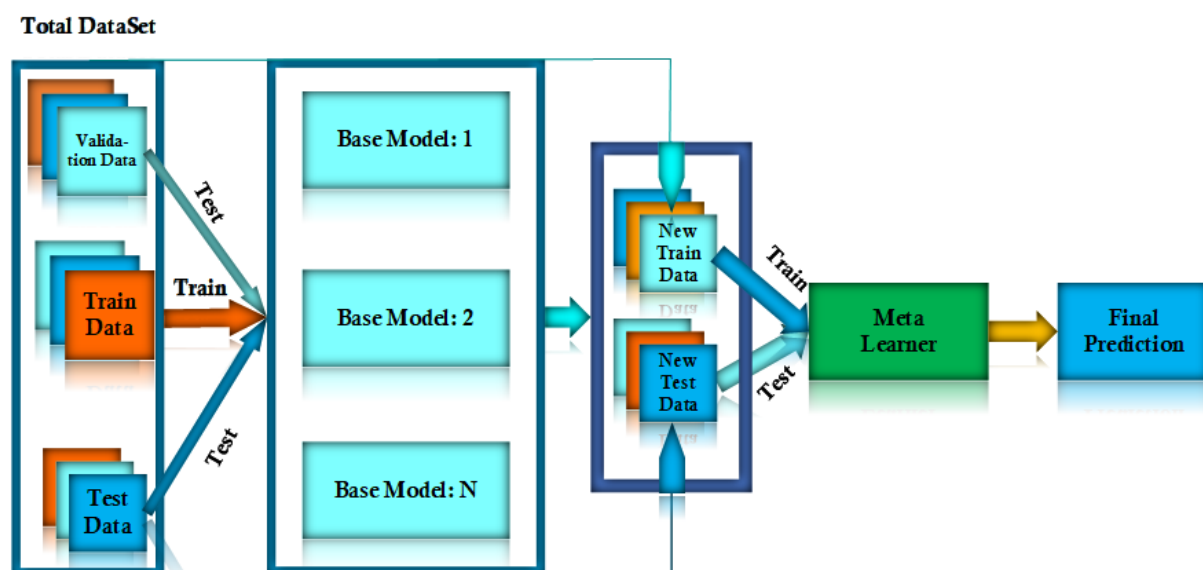


Figure 3.4.1 Staking Process

Predictions on the Validation Set: Once the base models have been trained, they are used to make predictions on the validation set. The next stage is to develop a meta-model, also known as a meta learner, which will take the predictions of the underlying models as input and make the final prediction. Any algorithm, such as linear regression, logistic regression, or even a neural network, can be used to create this model.

Training the Meta Model: The meta-model is then trained using the predictions given by the base models on the validation set. The base models' predictions serve as features for the meta-model.

Making Test Set Predictions: Finally, the meta-model is used to produce test set predictions. The basic models' predictions on the test set are fed into the meta-model, which then makes the final prediction.

Model Evaluation: The final stage is to assess the stacking ensemble's performance. This is accomplished by comparing the stacking ensemble's predictions to the actual values on the test set using evaluation measures such as accuracy, precision, recall, F1 score, and so on.

In the end, the goal of stacking is to combine the strengths of various base models by feeding them into a meta-model, which learns how to weight and combine their forecasts to generate the final prediction. This can frequently result in higher performance than utilizing a single model alone.

A simple way to understand the stacking process:

Step 1: Split the training dataset into two parts:

Dataset	Description
Training Data	Used to train the base models
Hold-Out Validation Data	Used to train the meta-model

Step 2: Train several base models on the training data:

Model	Algorithm
Base Model 1	EfficientNetB0 Model
Base Model 2	MobileNet Model

Step 3: Make predictions using the base models on the hold-out validation data:

Model	Base Model 1	Base Model 2
Prediction	Prediction 1	Prediction 2

Step 4: Train the meta-model on the hold-out validation data using the predictions from the base models as in:

Model	Algorithm
Meta-Model	Logistic Regression

Step 5: To make a prediction for new data:

Model	Algorithm
Base Model 1	EfficientNetB0 Model
Base Model 2	MobileNet Model
Meta-Model	Logistic Regression

Final step: Evaluate the performance of the stacked model on a separate test dataset that was not used during training.

The Advantages of Stacking:

Stacking has various advantages in machine learning:

Improved Predictive Performance: Stacking can reduce bias and variance in the final forecast by merging the results of numerous base models, resulting in improved predictive performance. Stacking allows the meta-model to learn from the capabilities of several base models and potentially capture intricate patterns that individual models may not be able to capture alone.

Model Diversity: Stacking promotes the usage of varied base models that may be trained using a variety of algorithms, architectures, and hyperparameter settings. This diversity can serve to lessen the danger of overfitting while also making the stacked ensemble more robust to different types of data.

Flexibility: Stacking is a versatile strategy that may be used to solve a variety of machine learning issues, including classification, regression, and even time series forecasting. It can be used with a variety of basic models, including decision trees, support vector machines, neural networks, and others.

Interpretability: Stacking can also reveal the significance of many base models and their forecasts for the final prediction. I can acquire a better grasp of their relative importance and interpretability by studying the weights or contributions of each base model in the meta-model.

The most often used Stacking method:

Simple stacking, also known as basic stacking or vanilla stacking, is the most frequent type of stacking used in machine learning. This is due to the fact that it is reasonably simple to implement and can produce good outcomes in many circumstances. The predictions of the underlying models are aggregated using a simple aggregation function, such as averaging or summing, in simple stacking. The aggregated forecasts are then utilized to train the meta-model, which makes the final prediction. Stacking with meta-features or stacking with distinct input features, on the other hand, can be successful in some cases. The sort of stacking you use will be determined by the specific problem you're attempting to answer, the features of your data, and the performance of the many models you're utilizing as base models.

3.4.2 Logistic Regression

Logistic regression is a type of regression model used for binary classification tasks, where the output variable is categorical. The model estimates the probability of an event occurring, typically between two classes, by applying a sigmoid function to a linear combination of input features.

Logistic Function: Logistic regression uses the sigmoid activation function to map any real-valued number into a probability between 0 and 1. The sigmoid function is defined as:

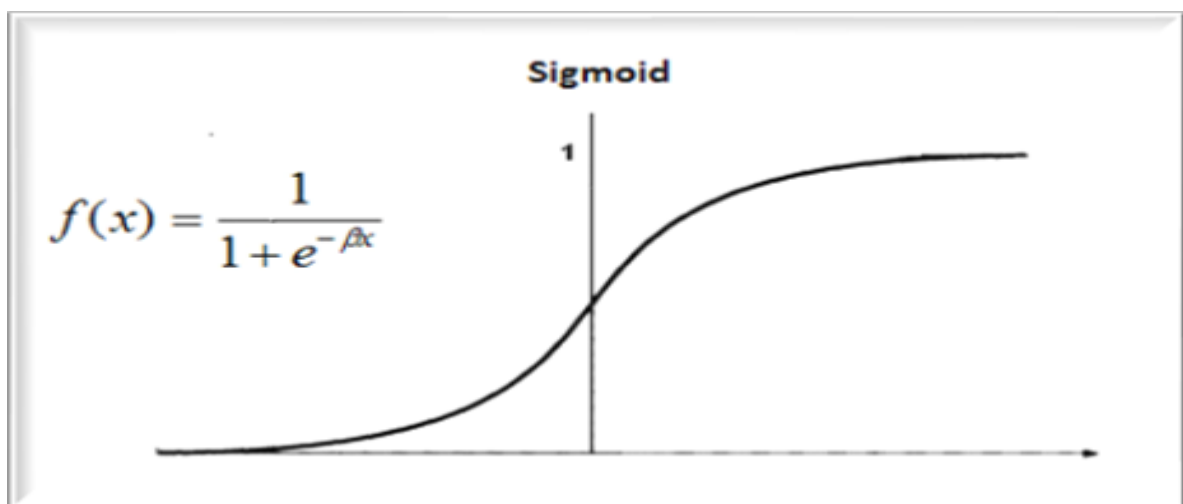


Figure: 3.4.2: Logistic Function

Here,

z is the weighted sum of the input features, and the output

$\sigma(z)$ represents the probability that the input belongs to the positive class (e.g., malignant tissue).

Classification: After the sigmoid function is applied, the output is interpreted as the predicted probability that an input belongs to one of the two categories (benign or malignant). If the probability is above a given threshold (typically 0.5), the input is classified into the positive class (malignant), otherwise, it is classified as negative (benign).

Training: The model is trained by optimizing the weights of the input features using maximum likelihood estimation, which minimizes the log-loss (cross-entropy loss). Logistic regression is efficient to train, even on large datasets, and is less prone to overfitting compared to more complex models, particularly when used with regularization techniques like L2 (Ridge) regularization. While logistic regression is limited in its ability to capture non-linear patterns, its simplicity and interpretability make it a useful tool for binary classification problems when integrated into more complex deep learning architectures.

In breast cancer histography analysis, pre-trained deep learning models (such as EfficientNetB0Net, MobileNet, or ResNet) are typically used to extract features from histopathological images. Logistic regression is then used as the final classification layer to make the binary decision—whether the input image represents benign or malignant tissue.

Convolutional neural networks (CNNs) are highly effective in analyzing histopathological images due to their ability to capture spatial hierarchies of features. Pre-trained models such as VGG16, ResNet, and EfficientNetB0Net are commonly used as feature extractors, leveraging transfer learning to adapt to breast cancer histopathology tasks. These models are pre-trained on large datasets (e.g., ImageNet) and can extract both low-level and high-level features from histopathological images, such as tissue texture, cell morphology, and nuclei patterns. Once the CNN extracts a feature representation from the input image, the output is a high-dimensional feature vector. However, CNNs alone do not produce binary classifications directly. This is where logistic regression comes into play.

Role of Logistic Regression in the Pre-Trained Model Pipeline:

In the deep learning pipeline for breast cancer histography analysis, logistic regression serves as the final classification layer that takes the output from the CNN feature extractor and produces a binary classification—benign or malignant. Here's how logistic regression integrates into the pipeline:

CNN Feature Extraction and Prediction: The pre-trained deep learning model processes the input histopathological images and outputs a prediction for meta model training after make feature vector representing the learned patterns in the image data.

Logistic Regression as the Classification Layer: The feature vector from the CNN is passed to a logistic regression model, which computes a weighted sum of the features, applies the sigmoid function, and outputs the probability that the image belongs to the malignant class.

Training: The pre-trained CNN can be fine-tuned on breast cancer histopathological datasets, while the logistic regression layer is trained from scratch using the extracted features.

The entire model is typically trained end-to-end, with the weights of both the CNN and logistic regression being optimized using backpropagation and gradient descent.

In breast cancer histography analysis, stacked ensemble learning involves training multiple base learners (such as different CNN architectures like ResNet, MobileNet, and EfficientNetB0Net) and then using a logistic regression model to combine their outputs. The outputs from the base learners can be the predicted class probabilities or the extracted features from each model. The logistic regression model serves as a meta-classifier that learns how to best combine the predictions from the base models to produce the final classification. This approach has been shown to improve classification accuracy by leveraging the strengths of each base model and reducing the variance in predictions.

In a recent study on breast cancer histography analysis, researchers used an ensemble of pre-trained CNN models (ResNet, MobileNet, and EfficientNetB0) combined with logistic regression as the final classifier. The CNNs are pre-trained on the ImageNet dataset and fine-tuned on the BreakHis dataset, which contains histopathological images of breast tissue. Logistic regression was employed as the classification layer to combine the features extracted by the CNNs. The study showed that the ensemble model with logistic regression achieved a classification accuracy of 95%, significantly outperforming individual models. The simplicity and interpretability of logistic regression made it easy to understand the contribution of each base model in the ensemble. Logistic regression plays a crucial role in pre-trained deep learning models for breast cancer histography analysis. By serving as the final classification layer, logistic regression enables deep learning models to produce binary classifications while maintaining simplicity and interpretability. When used as a meta-classifier in ensemble models, logistic regression can combine the strengths of multiple pre-trained CNNs, leading to improved classification accuracy and generalization.

3.4.3 Training Process of the Proposed Model

- Split Data set into train, test and validation set.
- Base models train on train set.
- Get predictions from base models on validation set.
- Combined all predictions from base models and made new train set.
- Train meta model on new train set.
- Test base model on test data get prediction and combined.
- Test meta model with combined test data.
- Made final predictions by meta model for test set

Visualization:

- using matplotlib to plot the training and validation accuracy over epochs.
- uses the confusion matrix function from scikit-learn to create a confusion matrix and the heatmap feature of Seaborn to display it.
- The code shows an entire workflow for developing, training, assessing, and visualizing a breast cancer histopathology image classification model based on an Ensembled approach, where used MobileNet and EfficientNetB0 as base model and Logistic Regression as meta model for final classification. Remember that this code offers a fundamental framework and can be improved further with hyperparameter tuning, more complex evaluation metrics, and other upgrades.

Chapter 4: Result and Analysis

4.1 Model Evaluation Metrics

This section presents the metrics used to evaluate the performance of the proposed ensemble model and its individual base models. The focus is on standard metrics used in classification tasks, especially in the medical imaging domain, where the consequences of misclassification can be significant.

4.1.1 Confusion Matrix Analysis

The confusion matrix provides a detailed breakdown of the model's performance by showing how many instances are classified correctly and incorrectly for each class. This analysis is especially useful for identifying specific types of errors (false positives and false negatives) in medical image classification.

A typical confusion matrix for binary classification is structured as follows:

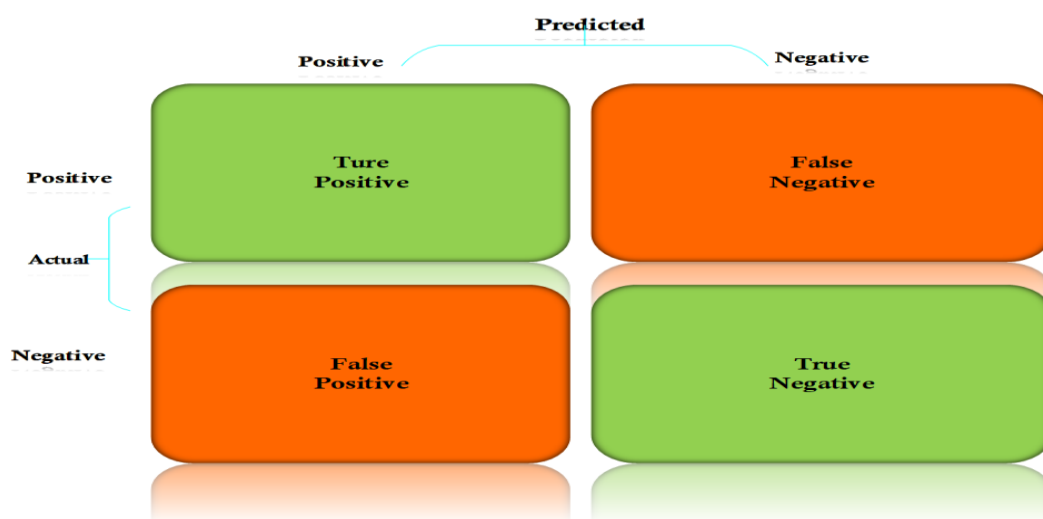


Figure 4.1.1: Confusion Matrix

True Negatives (TN): The number of benign samples correctly classified as benign.

True Positives (TP): The number of malignant samples correctly classified as malignant.

False Positives (FP): The number of benign samples incorrectly classified as malignant (Type I error).

False Negatives (FN): The number of malignant samples incorrectly classified as benign (Type II error).

The confusion matrix helps in identifying where the model struggles, such as whether it tends to misclassify malignant cases as benign (which is more dangerous in a medical context) or vice versa. By analysing the confusion matrix, you can better understand the model's behaviour and make adjustments to improve its sensitivity or specificity. These evaluation metrics provide a thorough and comprehensive way of analysing the performance of the proposed ensemble model for breast cancer histopathology, ensuring that the model is not only accurate but also reliable in distinguishing between benign and malignant cases.

4.1.2 Accuracy, Precision, Recall, F1-Score, Macro Average, weighted Average:

These core classification metrics are used to assess the model's overall performance, and each provides different insights into the quality of the predictions:

Accuracy: Accuracy measures the proportion of correctly classified instances out of the total number of instances. It is a general indicator of model performance, but in medical domains with imbalanced classes, it may not always be the most informative metric.

$$\text{Accuracy} = \frac{\text{True Positives}(TP) + \text{True Negatives}(TN)}{\text{Total Instances}} \dots \dots \dots (1)$$

Precision: Precision, also known as Positive Predictive Value, is the proportion of true positive predictions out of all positive predictions made by the model. It is particularly useful when the cost of false positives is high.

$$\text{Precision} = \frac{\text{True Positives}(TP)}{\text{True Positives}(TP) + \text{False Positives}(FP)} \dots \dots \dots (2)$$

Recall (Sensitivity): Recall, or sensitivity, measures the proportion of actual positives that are correctly identified by the model. In medical diagnostics, this metric is crucial because it highlights the model's ability to detect positive cases (e.g., cancer).

$$\text{Recall} = \frac{\text{True Positives}(TP)}{\text{True Positives}(TP) + \text{False Negatives}(FN)} \dots \dots \dots (3)$$

F1-Score: The F1-Score is the harmonic mean of precision and recall, providing a single metric that balances both. It is particularly useful when there is an uneven class distribution or when both false positives and false negatives are important to minimize.

$$\text{F1-Score} = 2 * \frac{\text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}} \dots \dots \dots (4)$$

Macro Average: The macro-average computes the mean by averaging the precision, recall, and F1 scores for each class separately.

weighted Average: The weighted average takes into account how many samples are included in each class. These metrics offer a comprehensive evaluation of the model's performance, balancing the need to avoid both false positives and false negatives, which is critical in breast cancer diagnosis.

4.1.3 ROC Curve and AUC (Area Under the Curve)

The Receiver Operating Characteristic (ROC) curve and the Area Under the Curve (AUC) are used to evaluate the model's ability to distinguish between positive and negative classes across different threshold values.

ROC Curve: The ROC curve plots the true positive rate (sensitivity) against the false positive rate at various classification thresholds. The curve illustrates the trade-off between recall and false positive rate, showing how the model's performance varies as the threshold changes.

AUC (Area Under the Curve): AUC quantifies the overall performance of the model by measuring the area under the ROC curve. A model with an AUC of 1 is perfect, while an AUC of 0.5 suggests the model performs no better than random guessing. A higher AUC indicates better performance in distinguishing between benign and malignant samples, making it an essential metric for evaluating classification models in medical imaging.

4.2 Performance of the Base Models

4.2.1 Performance Metrics for the First Base Model EfficientNetB0:

Confusion Matrix of Base Model EfficientNetB0:

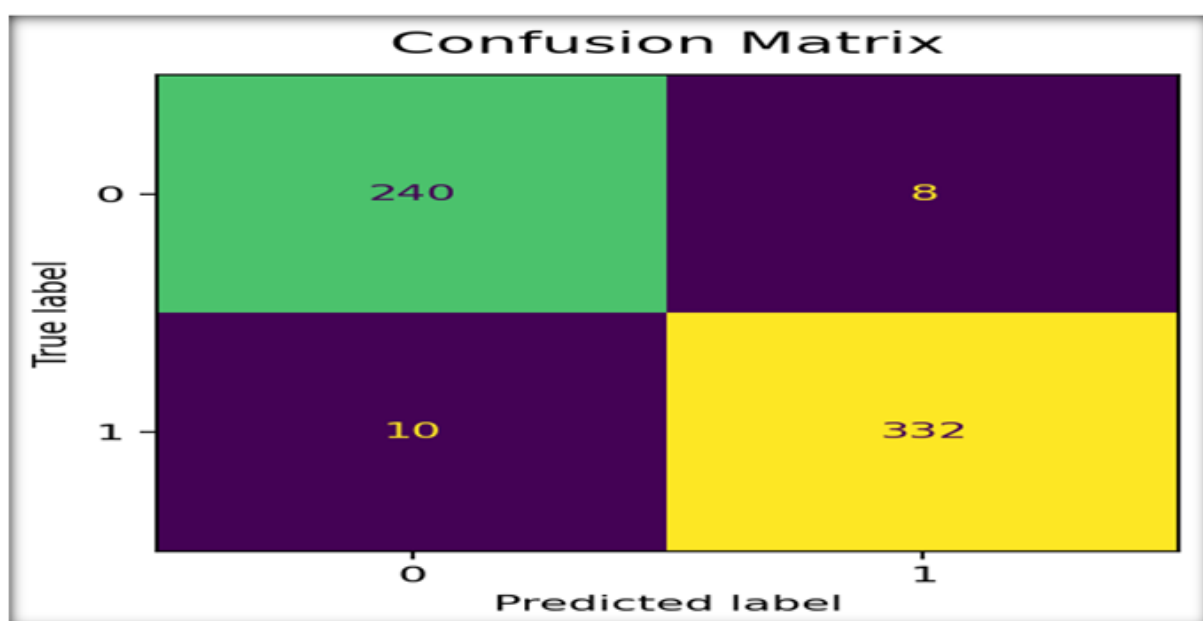


Figure 4.2.1: Confusion Matrix of Efficient Model

- The model performs admirably overall, with an accuracy of roughly 96.6%
- The model is doing Ill in both classes ("benign" and "malignant"), as evidenced by its excellent precision score (for benign=0.96 and malignant=0.97), recall (for benign=0.96 and malignant=0.97), F1-score (for benign=0.96 and malignant=0.97) with also another score which is support and this for benign is 248 and for malignant is 342.
- This over all accuracy for precision=0.96, recall=0.96 and f1=0.96 with support 0.969
- It uses a macro average to correct for class imbalances while still producing a high score of roughly 0.96 for precision, 0.969 for recall and 0.969 for f1-score with support 590.00.
- It uses also a weighted average to correct for class imbalances while still producing a high score of roughly for precision is 0.969, recall 0.969 and f1-score 0.969 with support 590.00.
- According to the assertion, the model exhibits a balanced trade-off between recall and precision, pointing to strong overall classification performance.

Evaluation Matrix of Base Model MobileNet:

	Metric	Value
0	ROC-AUC	0.997
1	Accuracy	0.966
2	Loss	0.076

Classification Report for sequential_2

	precision	recall	f1-score	support
0	0.960000	0.967742	0.963855	248.000
1	0.976471	0.970760	0.973607	342.000
accuracy	0.969492	0.969492	0.969492	0.969
macro avg	0.968235	0.969251	0.968731	590.000
weighted avg	0.969547	0.969492	0.969508	590.000

Figure: 4.2.1: Evaluation Matrix of EfficientNetB0 Model

Train and Validation Accuracy Curve of Base Model EfficientNetB0:

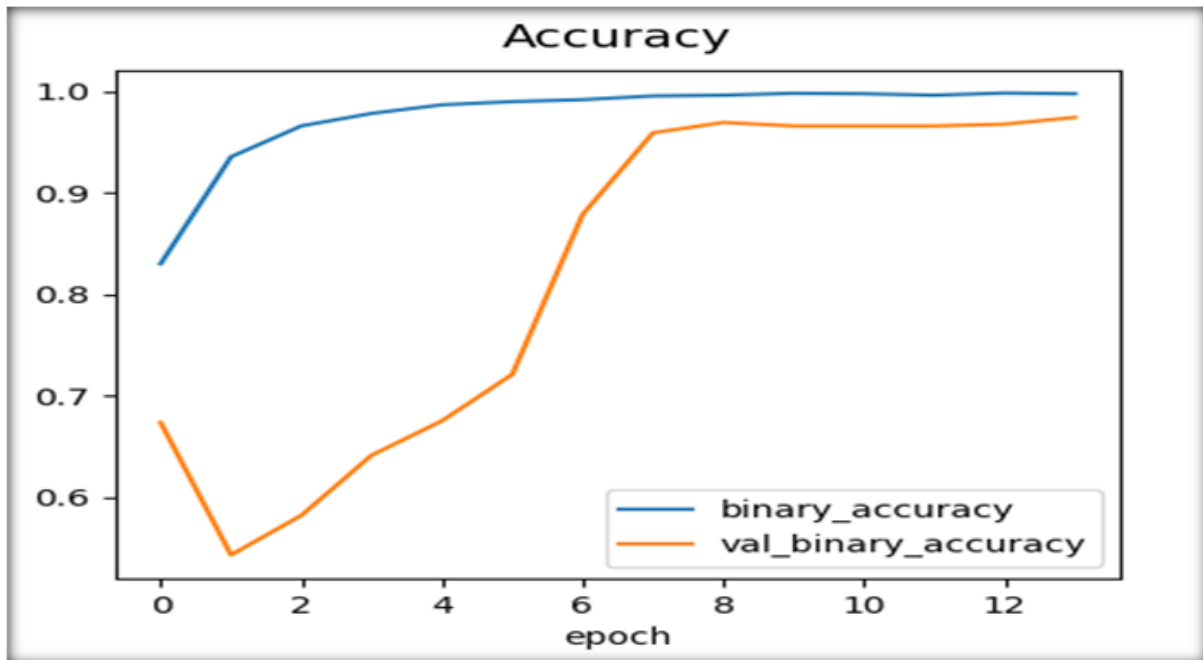


Figure 4.2.1: Train-Validation Accuracy Curve of EfficientNetB0 Model

Train and Validation Loss Curve of Base Model EfficientNetB0:

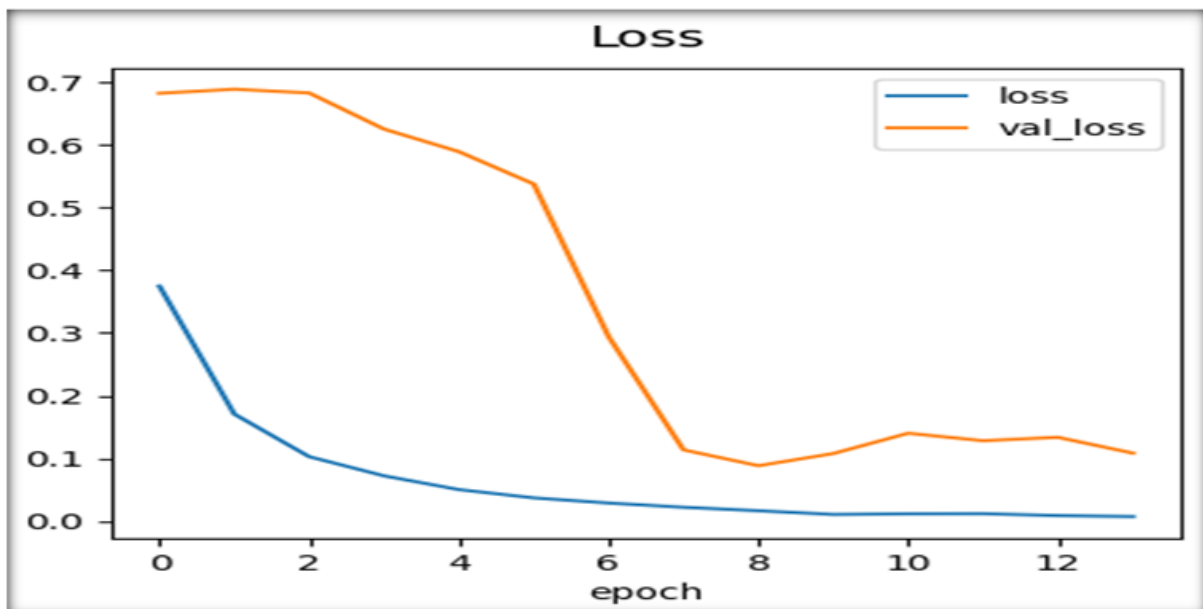


Figure 4.2.1: Train-Validation Loss Curve of EfficientNetB0 Model

Precision-Recall Curve of Base Model EfficientNetB0:

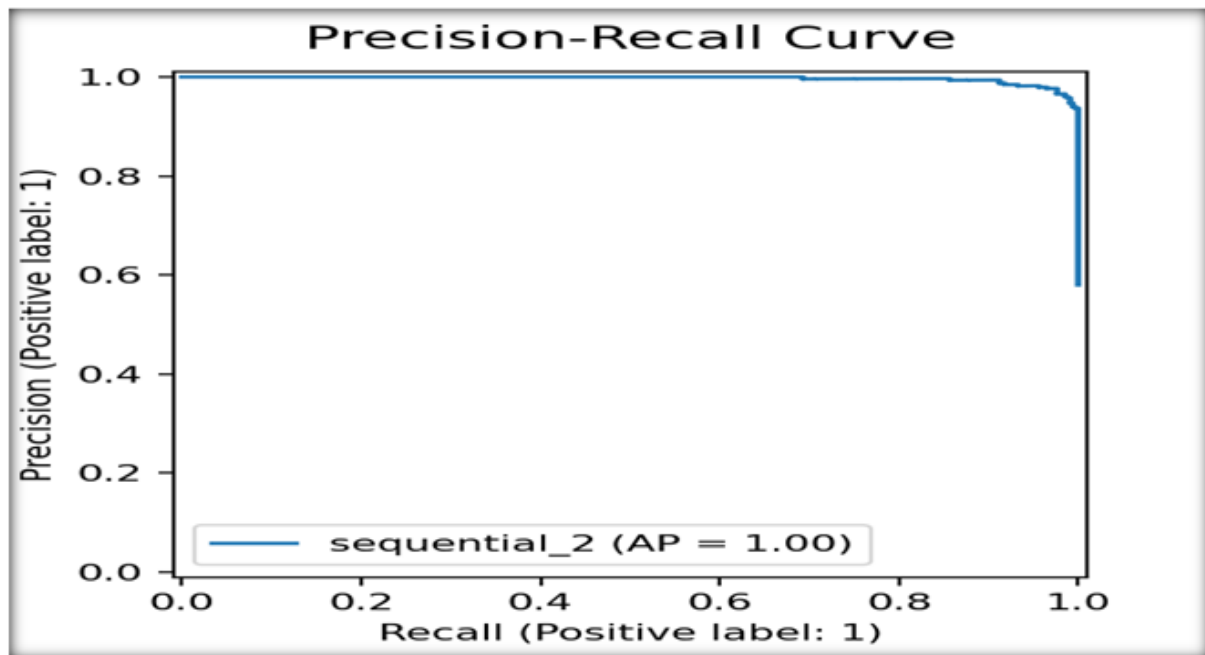


Figure 4.2.1: Precision-Recall Curve of EfficientNetB0 Model

ROC Curve and AUC (Area Under the Curve) of EfficientNetB0 Model:

From Evaluation Matrix Figure I can see the ROC-AUC value of the base model EfficientNetB0 is 0.997

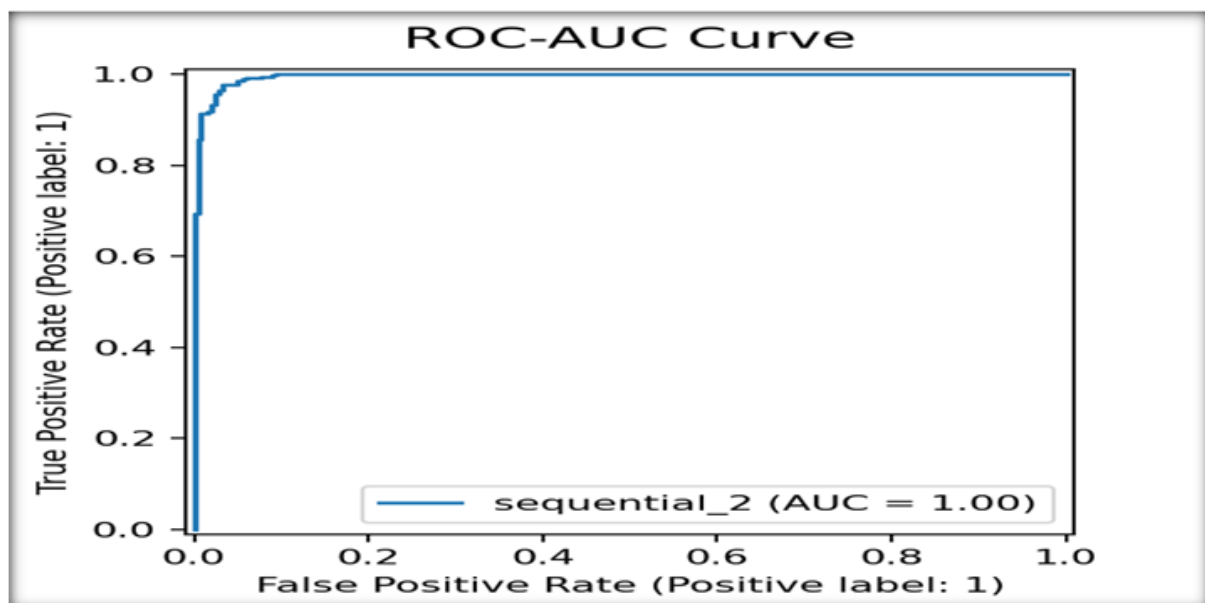


Figure 4.2.1: ROC-AUC Curve of Base Model EfficientNetB0

4.2.2 Performance Metrics for the Second Base Model MobileNet:

Confusion Matrix of Base Model MobileNet:

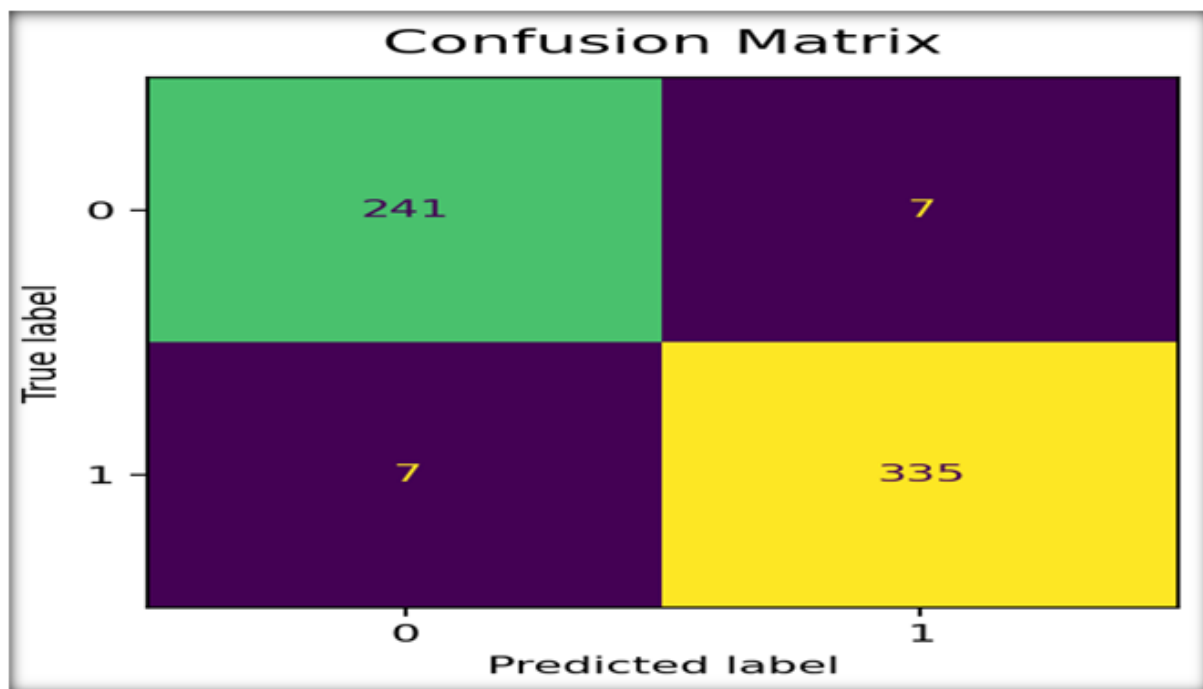


Figure 4.2.2: Confusion Matrix of MobileNet Model

- The model performs admirably overall, with an accuracy of roughly 97.6%.
- The model is doing ill in both classes ("benign" and "malignant"), as evidenced by its excellent precision score (for benign=0.97 and malignant=0.979), recall (for benign=0.97 and malignant=0.979), F1-score (for benign=0.97 and malignant=0.979) with also another score which is support and this for benign is 248 and for malignant is 342.
- This over all accuracy for precision=0.97, recall=0.97 and f1=0.97 with support 0.97.
- It uses a macro average to correct for class imbalances while still producing a high score of roughly 0.975 for precision, 0.975 for recall and 0.975 for f1-score with support 590.00.
- It uses also a weighted average to correct for class imbalances while still producing a high score of roughly 0.976. where for precision is 0.976, recall 0.976 and f1-score 0.976 with support 590.00.
- According to the assertion, the model exhibits a balanced trade-off between recall and precision, pointing to strong overall classification performance.

Evaluation Matrix of Base Model MobileNet:

	Metric	Value
0	ROC-AUC	0.996
1	Accuracy	0.976
2	Loss	0.073

Classification Report for sequential_1				
	precision	recall	f1-score	support
0	0.971774	0.971774	0.971774	248.000
1	0.979532	0.979532	0.979532	342.000
accuracy	0.976271	0.976271	0.976271	0.976
macro avg	0.975653	0.975653	0.975653	590.000
weighted avg	0.976271	0.976271	0.976271	590.000

Figure: 4.2.2 Performance Matrix of Base Model MobileNet

Train and Validation Accuracy Curve of Base Model MobileNet:

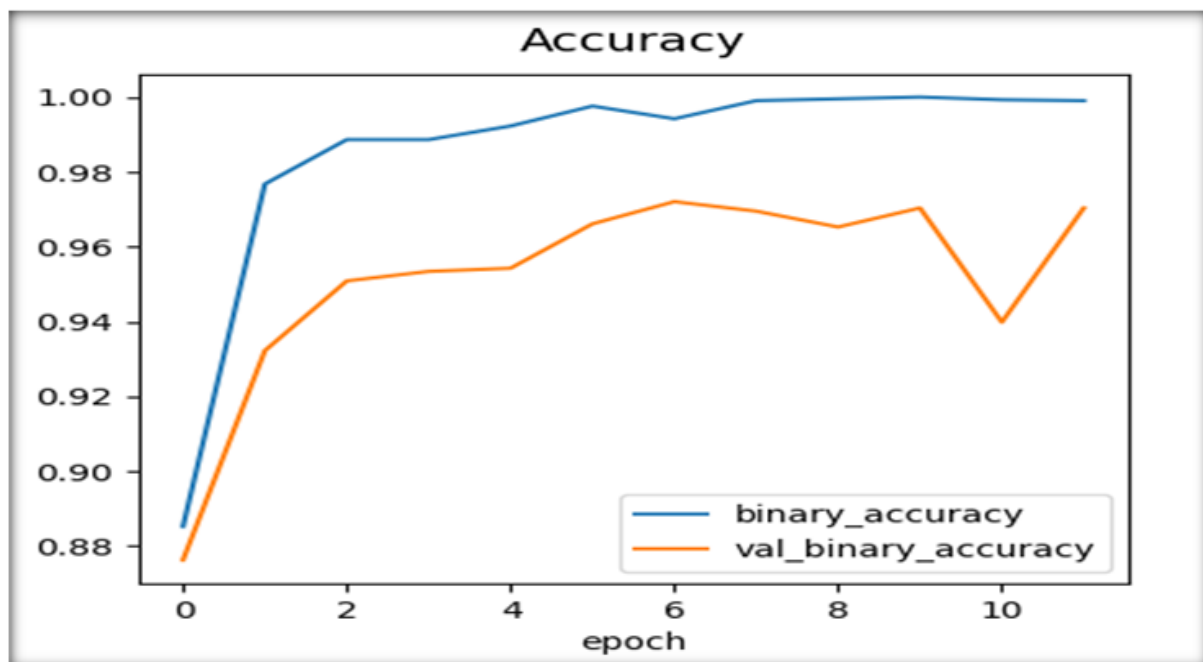


Figure 4.2.2: Train-Validation Accuracy Curve of MobileNet Model

Train and Validation Loss Curve of Base Model MobileNet:

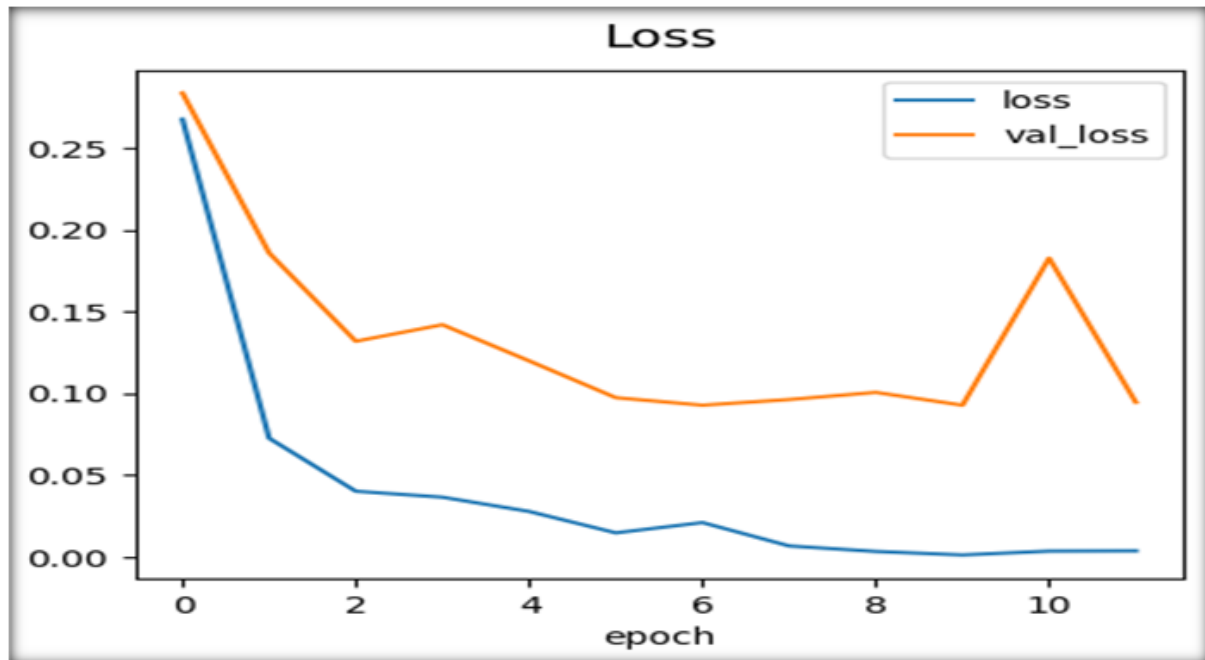


Figure 4.2.1: Train-Validation Loss Curve of MobileNet Model

Precision-Recall Curve of Base Model MobileNet:

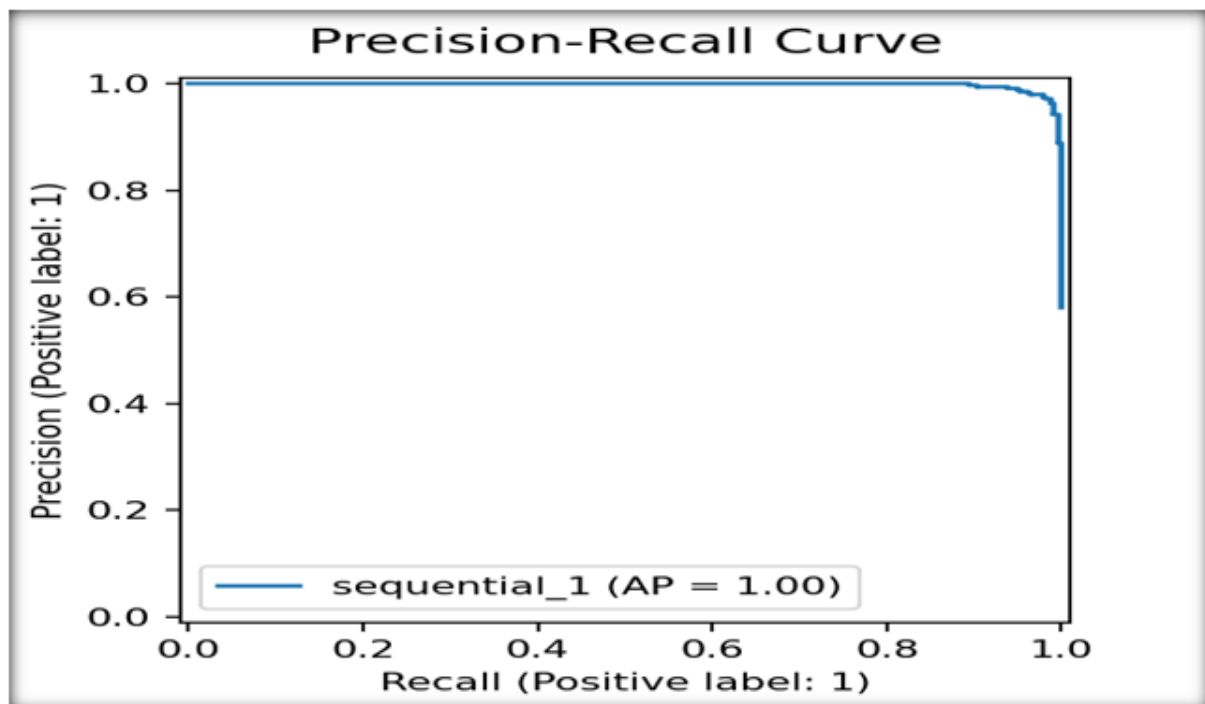


Figure 4.2.2: Precision-Recall Curve of MobileNet Model

ROC Curve and AUC (Area Under the Curve) of MobileNet Model:

From Evaluation Matrix Figure I can see the ROC-AUC value of the base model MobileNet is 0.996.

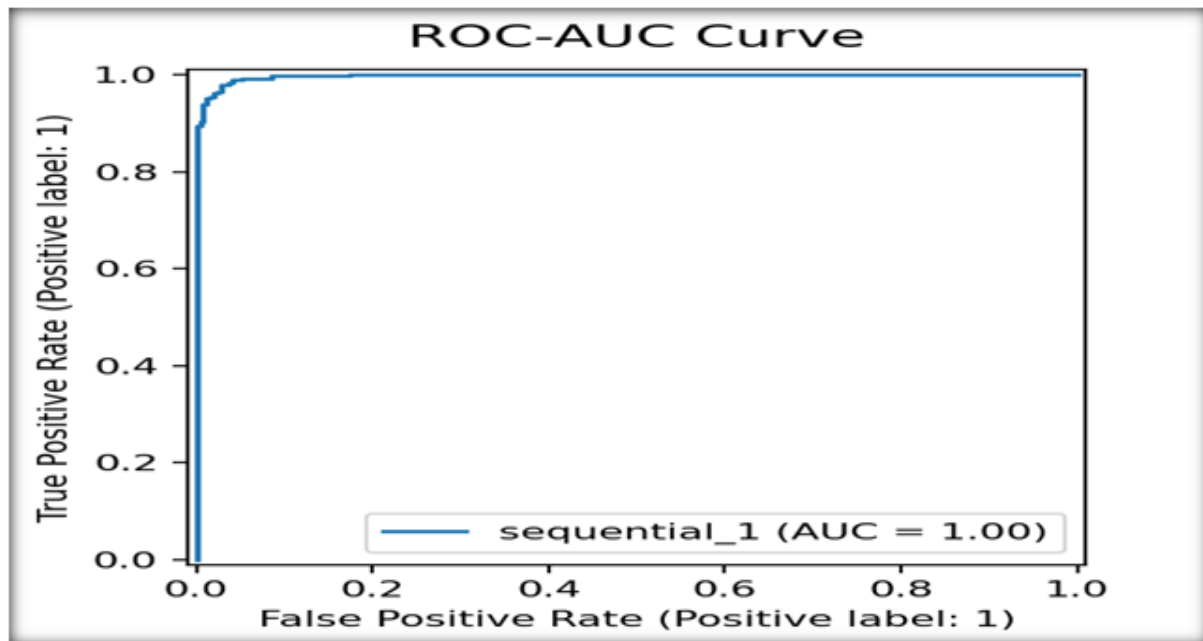


Figure 4.2.2: ROC-AUC Curve of Base Model MobileNet

Features Extraction of Base Models:

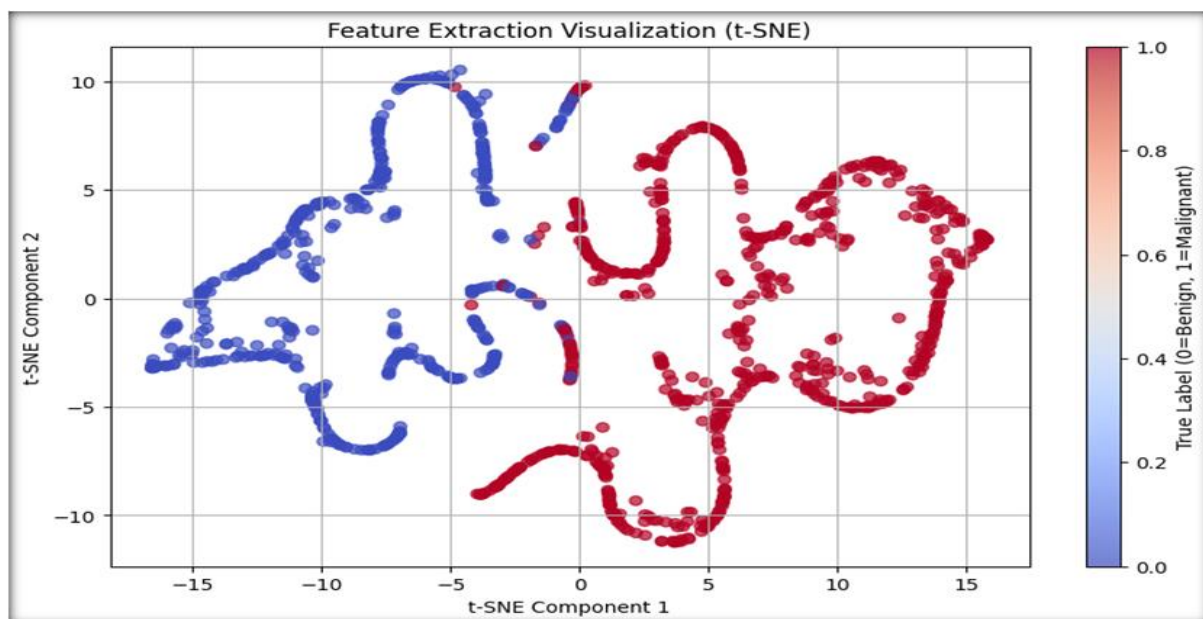


Figure 4.2.2 Feature Extraction of Base Models

4.3 Performance of Meta Model on Stacked Ensemble Approach

Confusion Matrix of Meta Model (Logistic Regression):

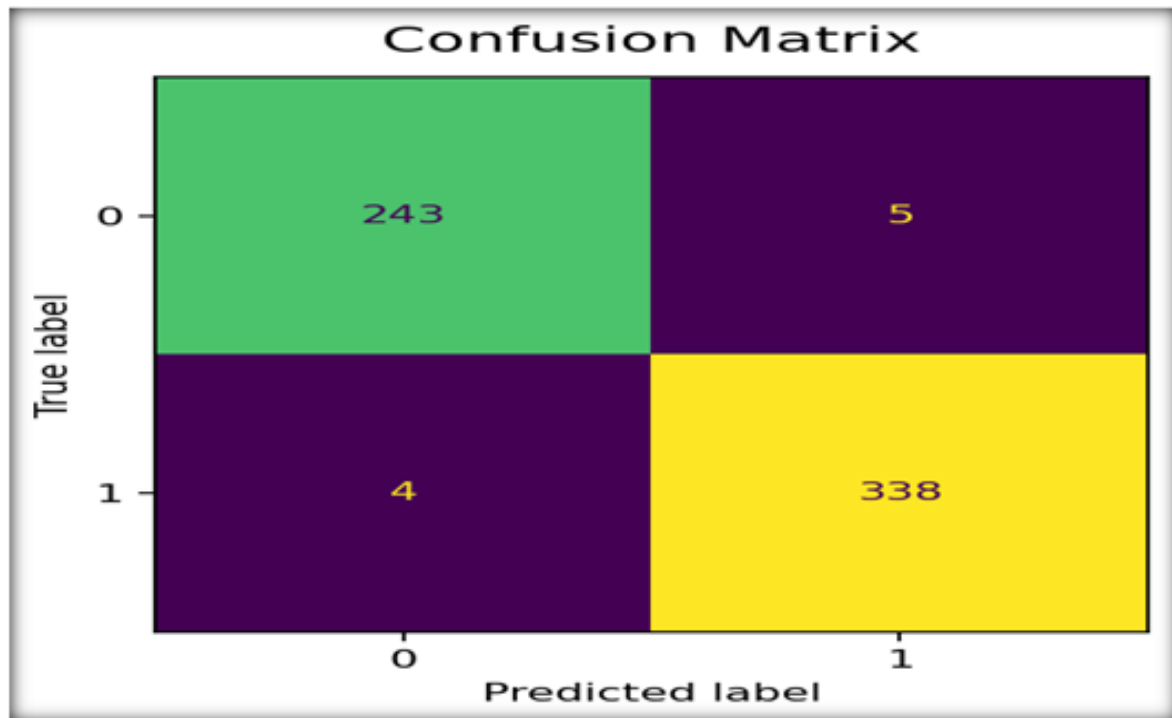


Figure 4.3: Confusion Matrix of Meta Model

- The model performs admirably overall, with an accuracy of roughly 98.5%.
- The model is doing ill in both classes ("benign" and "malignant"), as evidenced by its excellent precision score (for benign=0.98 and malignant=0.98), recall (for benign=0.97 and malignant=0.98), F1-score (for benign=0.98 and malignant=0.98) with also another score which is support and this for benign is 248 and for malignant is 342.
- This over all accuracy for precision=0.984, recall=0.984 and f1=0.984 with support 0.978.
- It uses a macro average to correct for class imbalances while still producing a high score of roughly 0.984 for precision, 0.984 for recall and 0.984 for f1-score with support 590.00.
- It uses also a weighted average to correct for class imbalances while still producing a high score of roughly 0.984. where for precision is 0.984, recall 0.984 and f1-score 0.984 with support 590.00.
- According to the assertion, the model exhibits a balanced trade-off between recall and precision, pointing to strong overall classification performance.

Evaluation Matrix of Meta Model (Logistic Regression):

	Metric	Value		
0	ROC-AUC	0.999		
1	Accuracy	0.985		
2	Loss	0.047		
Classification Report for Logistic Regression Stacking Model				
	precision	recall	f1-score	support
0	0.983806	0.979839	0.981818	248.000
1	0.985423	0.988304	0.986861	342.000
accuracy	0.984746	0.984746	0.984746	0.985
macro avg	0.984614	0.984071	0.984340	590.000
weighted avg	0.984743	0.984746	0.984741	590.000

Figure: 4.3 Performance Matrix of Meta Model

Precision-Recall Curve of Meta Model:

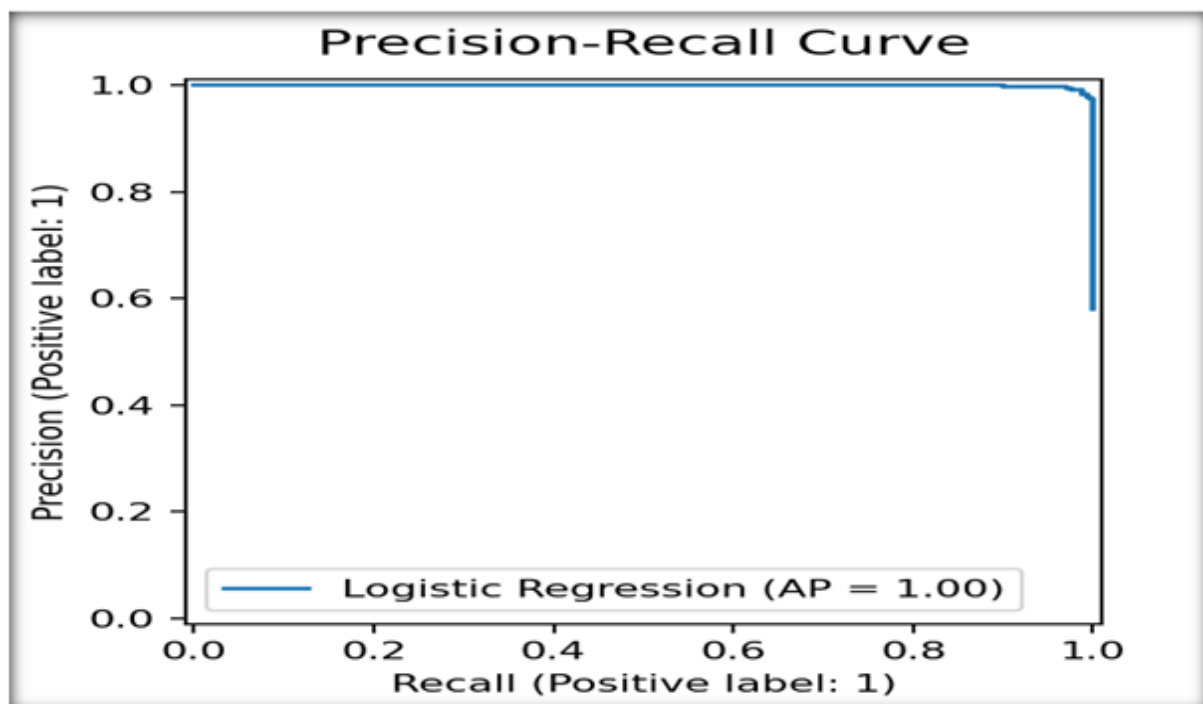


Figure 4.3: Precision-Recall Curve of Meta Model

ROC Curve and AUC (Area Under the Curve) of Meta Model:

From Evaluation Matrix Figure I can see the ROC-AUC value of the base model MobileNet is 0.999.

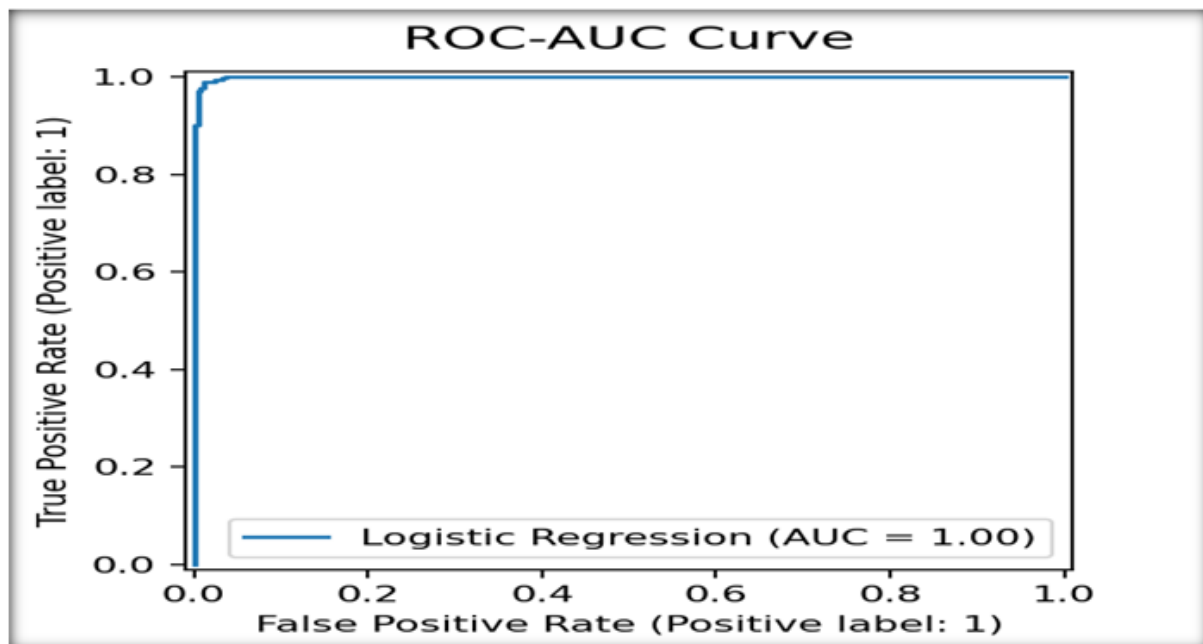


Figure 4.3: ROC-AUC Curve of Meta Model

4.4 Comparative Analysis

Comparative Performance of Base Models:

Base 1: EfficientNetB0 Base 2: MobileNetV1		
Loss	0.080017	0.065293
ROC-AUC	0.994559	0.996587
Accuracy	0.967797	0.977966

Figure 4.4 Comparative Performance of Base Models

Comparison of the performance Between Base Models and Meta Model:

	Base 1: EfficientNetB0	Base 2: MobileNetV1	Stacking Ensemble
Loss	0.080017	0.065293	0.046821
ROC-AUC	0.994559	0.996587	0.999186
Accuracy	0.967797	0.977966	0.984746

Figure 4.4 Performance Comparison Between Base Models and Meta Model

4.5 Comparison with Previous work on BreakHis Dataset.

Models	Approach	Accuracy (%)
Zhongyi Han [40] (2017)	Proposed Deep Learning Model	93.2%
Abdullah-Al Nahid [41] (2018)	DNN	91%
Majid Nawaz [42] (2018)	DenseNet CNN Model	95.4%
Pinky Agarwal, Anju Yadav [43] 2021)	VGG16	94.67%
Manisha Saini, Seba Susan [44] (2022)	VGG16	97.10%
Shiksha Singh, Rajesh Kumar [45] (2022)	Inception and ResNet	86.5%
Irem Sayin, Muhammed Ali Soydas [46] (2023)	Xception	89%
Proposed Stacked Approach in this Research.		98.5%

4.6 Final Finding of Proposed Model:

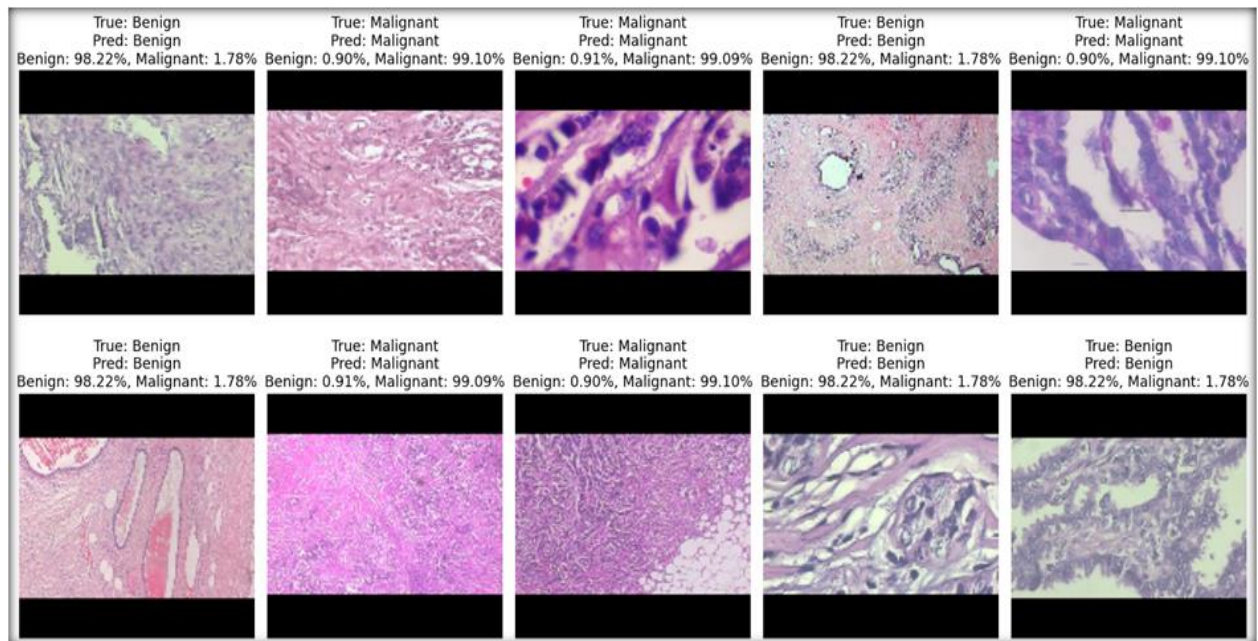


Figure 4.6: Model Final Performance

4.7 Code Implementation Process:

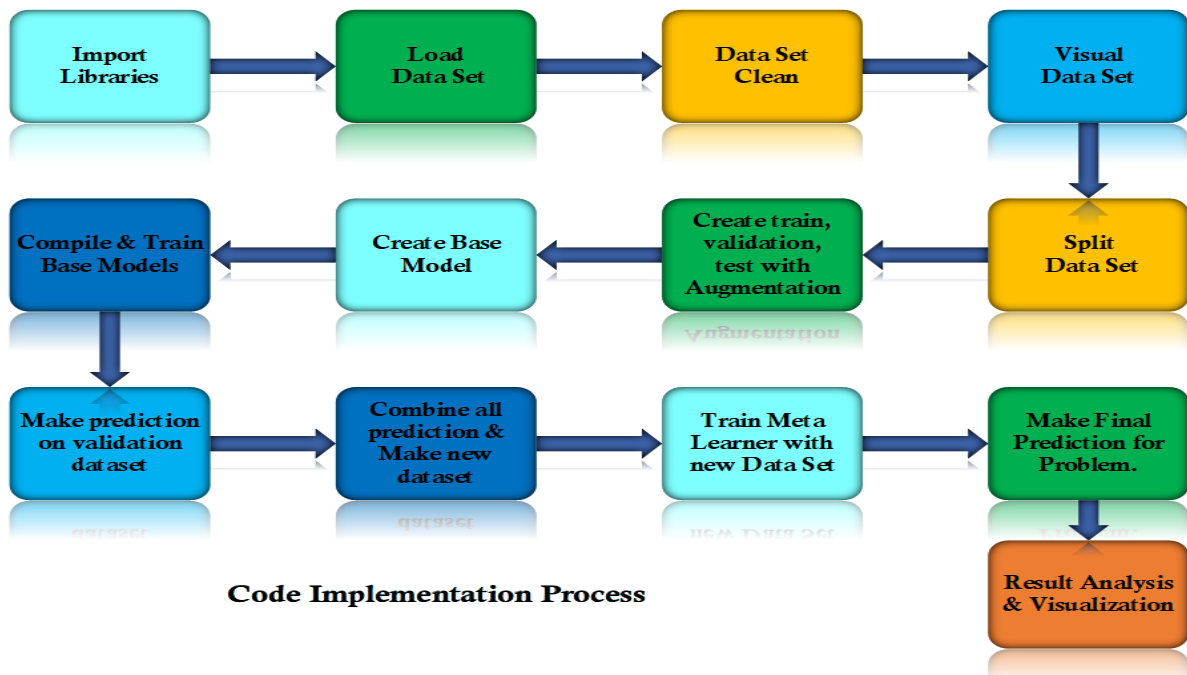


Figure 4.7: Coding Process

Chapter 5: Conclusion

5.1 Conclusion

A significant and fruitful line of investigation in the realm of medical image analysis and healthcare is the classification of breast cancer histopathology images using the Proposed Approach. Breast cancer histopathology image categorization using deep learning, more specifically the MobileNet and EfficientNetB0 architecture, has shown incredible promise for increasing the precision and effectiveness of cancer detection. It is the most prevalent cancer in America, where women are anticipated to have up to 30% more diagnoses of it in 2019[13]. In histopathology photos, the deep convolutional neural network MobileNet and EfficientNetB0 model have demonstrated outstanding accuracy in differentiating between malignant and benign breast tissue. And the accuracy of Proposed Stacking method is better than a single deep learning model. It is a good choice for this job because it can capture fine details in the photographs. As base models in Stacking MobileNet and EfficientNetB0 model reduces human interpretation's subjectivity by automating the classification process. And as meta model Logistic Regression best for final prediction or classification. Although radiologists' extensive knowledge is a major factor in their ability to successfully identify malignant tumors from histopathological pictures, specialists occasionally disagree with their conclusions. A second option for picture diagnosis is provided by computer-aided diagnosis, which can increase the accuracy of experts' judgment. Large datasets can be used to train deep learning models like MobileNet and EfficientNetB0 model, making them flexible for a variety of populations and capable of identifying different cancer subtypes. For dealing with the variety of breast cancer cases, this scalability is essential. MobileNet, EfficientNetB0 model and comparable models can still be difficult to use in practical clinical settings, despite their promise. Continuous hurdles include ensuring robustness, addressing data privacy concerns, and integrating new models into the current healthcare infrastructure. MobileNet, EfficientNetB0 model and other deep learning models frequently lack transparency in their decision-making. It is necessary to make an effort to improve interpretability so that clinicians can understand the rationale behind a specific classification. That's why I tried to combined multiple deep learning models in a single model where I used Logistic Regression for final classification. MobileNet and EfficientNetB0 model success is highly dependent on the caliber and variety of the training data. It is important to make an effort to ensure that the training datasets are impartial and representative. Breast cancer histopathology image categorization using proposed method holds significant promise for enhancing the precision, effectiveness, and objectivity of cancer diagnosis. Even if there are still issues and concerns, more research and advancement in this area have the potential to have a big influence on early breast cancer diagnosis and treatment, leading to better patient outcomes. In this study, our proposed architecture was rated as the best model for performance and accuracy in the identification and classification of breast cancer using transfer learning. Stacking method is a wonderful and effective predictor in the field of health [15].

5.2 Future Study

It is accepted that the pursuit of a flawless, 100% accurate output through fine-tuning is an admirable goal in the context of this undertaking. The current endeavor, however, is unable to do the thorough investigation and improvement necessary to achieve such a degree of precision due to time restrictions. As a result, it is anticipated that future research projects will offer the chance to devote the required amount of time and money to this effort, potentially leading to superior results in terms of accuracy and performance.

5.2.1 Recommendations for Future Research

To bridge the identified gaps, future research should consider the following approaches:

- **Data Augmentation and Synthesis:** Utilize advanced data augmentation techniques and synthetic data generation (e.g., GANs) to expand and balance datasets.
- **Developing Explainable Models:** Invest in research on XAI techniques tailored for medical imaging to improve model transparency.
- **Cross-Dataset Validation:** Perform extensive validation across multiple datasets to assess model generalizability.
- **Multi-Modal Integration:** Explore the integration of various data types (imaging, genomic, clinical) to create comprehensive classification models.
- **Collaborative Efforts:** Encourage collaborations between data scientists, clinicians, and regulatory bodies to ensure models meet clinical needs and standards.
- **Standardization Initiatives:** Advocate for the development of standardized datasets, evaluation protocols, and benchmarking platforms to facilitate consistent and comparable research outcomes.

By addressing these recommendations, the research community can effectively overcome current limitations and advance the application of CNNs in breast cancer classification.

5.3 Limitations

Data Availability and Diversity: The effectiveness of the classification models heavily depends on the quality, diversity, and size of the dataset. Limited availability of high-quality labeled histopathology images could impact the generalization and performance of the models.

Data Imbalance: Imbalance among different subtypes and grades of breast cancer could lead to biased classification results, favoring the majority class. Addressing data imbalance is crucial to ensure fair and accurate predictions.

Complexity of Breast Cancer: Breast cancer is a complex disease with various subtypes and grades, making accurate classification challenging. Models might struggle with distinguishing subtle differences among closely related classes.

Deep Learning Models: While efforts will be made to enhance interpretability, deep learning models can still be challenging to interpret comprehensively. The exact decision-making process within the models might not be fully transparent.

Clinical Variability: The clinical relevance of the classification models might vary depending on factors such as the specific patient population, the availability of additional clinical data, and evolving medical guidelines.

Ethical Considerations: Implementing models in a clinical setting requires careful consideration of patient privacy, informed consent, and potential biases that could affect different demographic groups.

External Validation: Due to the unique characteristics of different medical institutions and patient populations, the developed models might need to be externally validated before widespread adoption.

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